To What Extent Do Magnetoplasmadynamic Thrusters Enable More Efficient and Faster Earth–Mars Transfer Trajectories Than the Hohmann Transfer Orbit?

1. Introduction

The trajectory between Earth and Mars is one of the most studied problems in mission design, yet it remains fundamentally constrained by the propulsion technologies available. For decades, the Hohmann transfer orbit has served as the canonical baseline: a minimum-energy ellipse connecting two planetary orbits using precisely two impulsive burns, coasting passively through the remainder of the journey. It is elegant, well-understood, and propellant-frugal within the specific logic of chemical propulsion. It is also slow, inflexible, and increasingly inadequate for any Mars mission architecture that prioritises crew safety, payload economy, or departure flexibility.

Magnetoplasmadynamic (MPD) thrusters represent a different propulsion paradigm entirely. Rather than delivering discrete velocity impulses, they accelerate plasma electromagnetically to produce continuous low thrust across mission durations of months. The resulting trajectories bear no resemblance to the fixed ellipses of classical orbital mechanics. They can be optimised dynamically, shaped by mission objectives, and extended across the full heliocentric arc. The theoretical advantages are well established. The question is whether those advantages survive contact with the engineering constraints that govern real spaceflight, particularly the demand for megawatt-class electrical power in a domain where the nearest power grid is 150 million kilometres away.

This essay investigates that question systematically. It establishes the Hohmann transfer as the reference case, examines the physical principles of MPD propulsion across its principal configurations, and analyses the nuclear-electric power systems that are its essential enablers. It then compares the trajectory design space opened by continuous thrust, including spiral escapes, optimised heliocentric transfers, brachistochrone profiles, and ballistic capture, against the Hohmann baseline. The human factors of radiation exposure and the theoretical possibility of acceleration-induced artificial gravity are also evaluated, followed by a frank assessment of the engineering limitations that bound the technology's current reach.

2. Fundamentals of Orbital Mechanics and the Hohmann Transfer Orbit

Orbital mechanics rests on a conceptually simple but practically powerful insight: once a spacecraft has been given a velocity, it follows a trajectory determined entirely by the gravitational field it inhabits. No propulsion is needed to sustain an orbit. Propulsion is needed only to change one, and every such change costs propellant. The Hohmann transfer, formulated by Walter Hohmann in 1925, identifies the minimum-energy path between two coplanar circular orbits using exactly two velocity impulses.

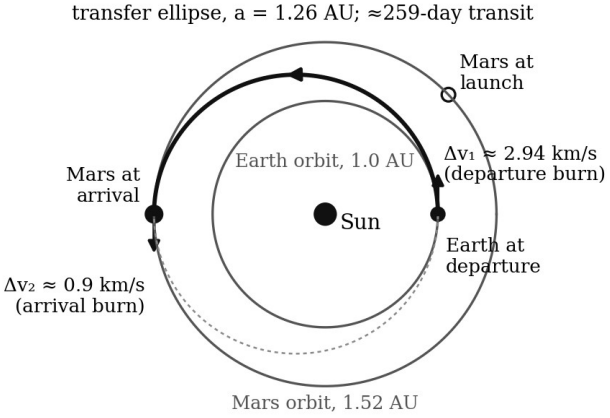


Fig. 1. Earth–Mars Hohmann transfer ellipse: two impulsive burns join the 1.0 AU and 1.52 AU orbits (schematic, not to scale).

For the Earth–Mars case, the geometry is determined by the orbital radii. Earth orbits at approximately 1.0 AU and Mars at approximately 1.52 AU. The semi-major axis of the transfer ellipse is the mean of these two values: (1.0 + 1.52) / 2 = 1.26 AU. By Kepler's third law, the orbital period of this ellipse scales as a^(3/2), giving a full period of approximately 1.415 years. The transit time occupies half of this ellipse, yielding a one-way transfer time of roughly 259 days, or approximately eight and a half months.

The total delta-v budget for a Hohmann Earth–Mars transfer is approximately 3.6 km/s, comprising a departure burn of roughly 2.94 km/s from low Earth orbit onto the transfer ellipse and an arrival burn of approximately 0.9 km/s to circularise into Mars orbit. These values represent the theoretical minimum for a two-impulse transfer and are among the most propellant-efficient strategies available within the framework of impulsive propulsion.

The propellant cost of any delta-v manoeuvre is governed by the Tsiolkovsky rocket equation:

$$\Delta v = ve \cdot \ln(m_0 / mf)$$

where ve is the effective exhaust velocity, m_0 is the initial mass of the spacecraft including propellant, and mf is the final mass after the burn. The logarithmic relationship is critical: doubling delta-v more than doubles the propellant mass fraction for a given exhaust velocity. For chemical propulsion systems with exhaust velocities around 4.4 km/s (corresponding to Isp ≈ 450 s), the 3.6 km/s Hohmann delta-v budget implies that roughly 55% of the spacecraft's departure mass must be propellant. This makes payload mass fraction extremely sensitive to any additional manoeuvres.

The Hohmann transfer also enforces strict launch window constraints. Earth and Mars align in the necessary geometry only once per synodic period of approximately 779 days, or roughly 26 months. Missing a window imposes a full synodic cycle penalty, a constraint with significant logistical implications for any sustained Mars programme. Furthermore, the Hohmann transfer is optimal only under specific assumptions: coplanar circular orbits, two impulsive burns, and no thrust during the transfer. Once continuous thrust becomes available, all three of these assumptions break down, and the Hohmann transfer ceases to be the uniquely optimal solution.

3. Principles of Magnetoplasmadynamic Thrusters

MPD thrusters operate on electromagnetic acceleration rather than combustion. Propellant gas is ionised to form a plasma between a central cylindrical cathode and a surrounding annular anode. A high-current discharge through this plasma generates a Lorentz body force, described by the vector relation F = J × B, where J is the current density and B is the magnetic field. The resulting force accelerates the plasma axially, ejecting it at high velocity to produce thrust.

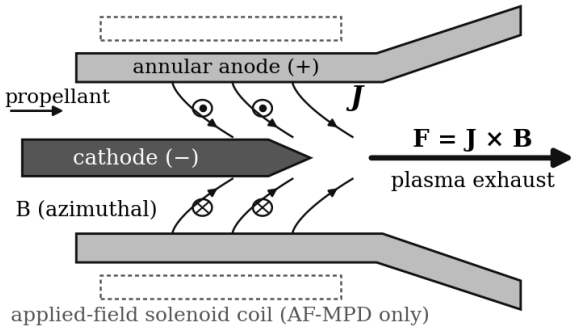


Fig. 2. Coaxial MPD thruster: radial current J and azimuthal field B produce the axial Lorentz force F = J × B; applied-field variants add an external solenoid (schematic).

The thrust F is related to the propellant mass flow rate and the exhaust velocity by:

$$F = \dot{m} \cdot ve$$

where $\dot{m}$ is the mass flow rate in kg/s and ve is the effective exhaust velocity in m/s. This relationship is exact and applies to all propulsion systems. The specific impulse, which measures the exhaust velocity in terms useful for propulsion engineering, is defined as:

$$Isp = ve / g_0$$

where $g_0 = 9.81$ m/s² is standard gravitational acceleration at Earth's surface. For a chemical rocket with Isp = 450 s, ve = 450 × 9.81 ≈ 4,415 m/s. For an MPD thruster operating at Isp = 5,000 s, ve = 5,000 × 9.81 = 49,050 m/s, more than eleven times higher. This difference transforms the rocket equation: the propellant mass required to deliver a given delta-v falls by orders of magnitude at high Isp.

Self-Field MPD Thrusters

Self-field devices rely entirely on the magnetic field generated by their own discharge current. This azimuthal self-field interacts with the radial current component to produce an axial Lorentz force. At discharge currents above approximately 10 kA, this mechanism becomes efficient, producing Isp values in the range of 2,000 to 6,000 seconds. A 150 kW superconducting device studied by Zheng et al. (2021) demonstrated a maximum thrust of 4 N and a specific impulse of 5,714 s, with a peak efficiency of 76.6%, representing the upper end of demonstrated MPD performance. However, electrode erosion at high current densities remains a critical lifetime constraint.

Applied-Field MPD Thrusters

Applied-field MPD thrusters supplement the self-generated field with an external solenoid coil. This allows efficient operation at lower discharge currents, reducing electrode erosion while improving performance stability. Recent work using high-temperature superconducting (HTS) magnets demonstrated thrust of 220 mN at 15 kW and Isp of 2,760 s, with efficiency improvements over conventional copper-wound equivalents (Aftab et al., 2024). The magnetic nozzle enhanced MPDT (ME-MPDT) variant has demonstrated a 20% thrust increase relative to standard applied-field designs at comparable input power (Wu et al., 2024).

Pulsed MPD Thrusters

Pulsed MPD thrusters address the power constraint differently. Instead of requiring sustained megawatt-class power, they charge capacitor banks over time and discharge in high-energy pulses that momentarily replicate the plasma conditions of much larger steady-state devices. This decouples peak operating conditions from average power availability. Sovey and Mantenieks (1991) reviewed a broad range of pulsed and steady-state MPD designs and found efficiencies of 0.43 and 0.69 at around 5,000 s Isp for hydrogen and lithium propellants respectively, with electrode power losses below 20% at megawatt power levels. The trade-off is lower average thrust and the additional system mass of the capacitor banks.

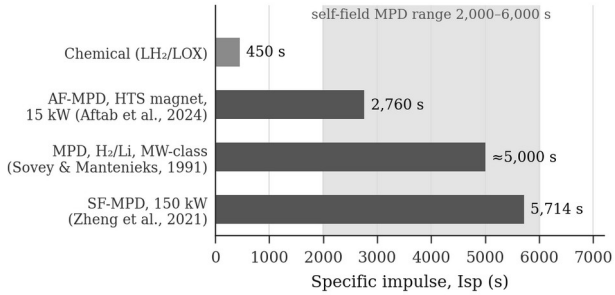


Fig. 3. Specific impulse of chemical propulsion and the MPD devices discussed in Section 3.

4. Nuclear-Electric Propulsion as an Enabling Constraint

Power is not merely an operational consideration for MPD thrusters; it is the defining constraint. Self-field MPD systems at mission-relevant scales require discharge powers in the hundreds of kilowatts to megawatt range. Solar power degrades inversely with the square of heliocentric distance. At Mars's mean orbital distance of 1.52 AU, solar irradiance is approximately 43% of its Earth-equivalent value. For a spacecraft requiring 500 kW of electrical power, a solar

array large enough to function near Earth becomes inadequate beyond the asteroid belt and impractical in terms of structure, mass, and drag during spiral escape manoeuvres.

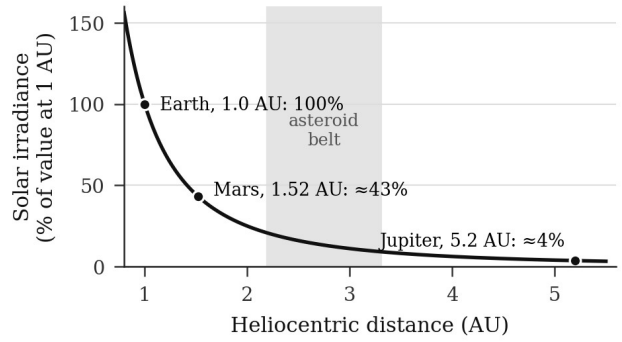


Fig. 4. Solar irradiance falls with the inverse square of heliocentric distance.

Nuclear-electric propulsion (NEP) is therefore not an optional enhancement but an enabling requirement for high-power MPD operation on deep-space missions. A NEP system uses a fission reactor to generate heat, which is converted to electricity through dynamic power conversion cycles. Work conducted under NASA's Mars Transportation Assessment Study evaluated concepts using supercritical CO_2 Brayton conversion combined with modified terrestrial microreactor technology, identifying this approach as suitable for crew and cargo Mars missions in the late 2030s or early 2040s (Mason et al., 2022). Yuan et al. (2024) modelled a 6 MWt sCO_2 Brayton system and found that reheating re-compress configurations achieve cycle efficiencies of up to 39.47%, with the most mass-efficient configuration delivering a ratio of 7.42 kg/kWe.

Frisbee et al. (1993) evaluated the SP-100 reactor for Mars cargo missions using MPD thrusters. Their system used three reactor modules delivering 1.7 MWe total with lithium-propellant MPD thrusters at Isp = 5,000 s, achieving a power system specific mass of 24.8 kg/kWe. The MPD thrusters operated at 60% electric-to-jet efficiency. At 24.8 kg/kWe, a 1 MWe system carries approximately 24,800 kg in reactor, shielding, power conversion, and thermal management mass alone. This mass penalty must be weighed against propellant savings when comparing total mission mass across propulsion options.

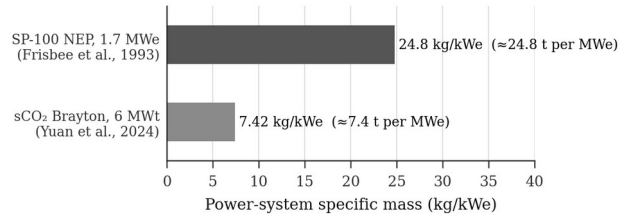


Fig. 5. Nuclear power-system specific mass for the two NEP studies discussed in Section 4.

Thermal management is particularly demanding. A reactor operating at 35% thermal efficiency must reject roughly 1.86 MW of waste heat for every 1 MW of electrical output. In space, the only heat rejection mechanism is thermal radiation, governed by:

$$P_{rad} = \varepsilon \cdot \sigma \cdot A \cdot T^4$$

where ε is emissivity, σ is the Stefan-Boltzmann constant (5.67×10^{-8} W/m²K⁴), A is radiator area, and T is radiator temperature. Operating at 1,000 K with ε ≈ 0.9, each square metre of radiator rejects approximately 51 kW. Rejecting 1.86 MW therefore requires roughly 36 m² of radiator surface. For multi-megawatt systems, required radiator areas can exceed several hundred square metres. Oleson et al. (2020) noted that even a 1.5 MWe reactor required radiator areas of approximately 2,500 m² for an opposition-class Mars mission, a significant structural and mass constraint.

5. Alternative Trajectories Enabled by Continuous Low-Thrust Propulsion

The availability of sustained thrust removes the fundamental constraint of classical trajectory design: that a spacecraft must follow a Keplerian orbit whenever no burn is active. This opens a design space qualitatively distinct from the two-impulse framework underlying the Hohmann transfer.

Spiral Escape Trajectories

Rather than performing a single high-thrust departure burn to achieve hyperbolic escape velocity, an MPD-propelled spacecraft can thrust continuously from low Earth orbit, gradually expanding its orbital radius in an outward spiral until it exceeds escape velocity.

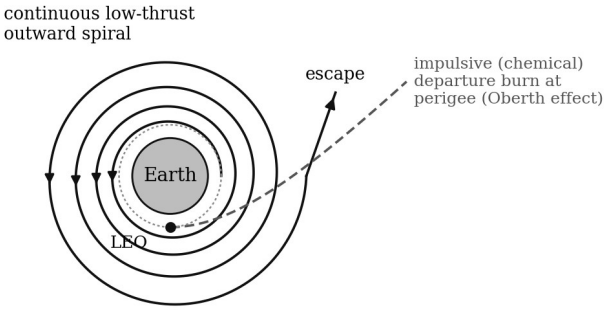


Fig. 6. Spiral escape trajectory from low Earth orbit compared with an impulsive departure (schematic, not to scale).

The spiral escape is energetically less efficient than a single impulsive burn, as continuous low-thrust departure sacrifices the Oberth effect, which states that propulsive manoeuvres are more efficient when performed at high orbital speed close to a massive body. Wall and Conway (2005) found that near-optimal low-thrust Earth–Mars trajectories using nuclear-electric propulsion require careful optimisation of the heliocentric thrust profile to compensate for this departure penalty. Despite the Oberth inefficiency, the propellant savings from high Isp more than offset this loss in propellant mass terms.

Continuous Heliocentric Low-Thrust Transfers

Once the spacecraft escapes Earth, it can apply continuous thrust throughout the heliocentric arc. Unlike the Hohmann ellipse, this trajectory is not fixed in shape. The spacecraft continuously accelerates or decelerates, following a path that depends on the optimisation objective. Morante et al. (2021) provide a comprehensive review of numerical approaches to this problem, noting that fuel-optimal and

time-optimal solutions generally require different thrust profiles and that the problem is computationally demanding due to the sensitivity of optimal co-states to initial conditions. At power levels of 200 to 500 kW, transit times for continuous heliocentric low-thrust transfers typically fall in the range of 200 to 270 days, a modest improvement over the 259-day Hohmann baseline.

Accelerate–Flip–Decelerate (Brachistochrone) Trajectories

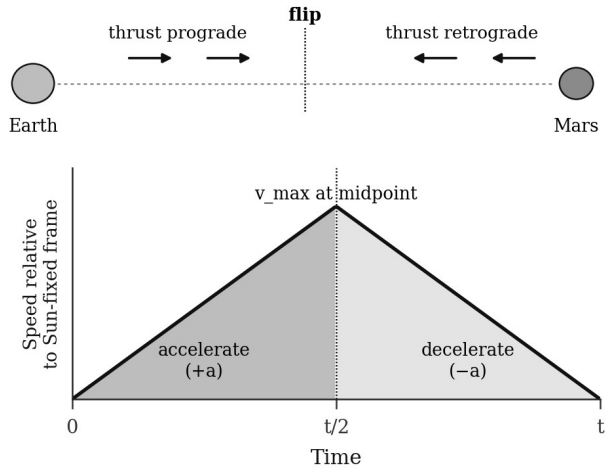


Fig. 7. Accelerate–flip–decelerate (brachistochrone) trajectory profile (schematic).

The brachistochrone trajectory is the most conceptually compelling of the continuous-thrust profiles. A spacecraft accelerates throughout the first half of the journey and decelerates during the second, arriving at Mars with near-zero relative velocity. The transit time scales with available acceleration. For constant acceleration a across a distance d, transit time t scales approximately as:

$$t \approx 2 \cdot \sqrt{(d / a)}$$

For the mean Earth–Mars distance of approximately 2.24×10^{11} m, a constant acceleration of 0.001 m/s² yields a transit time of roughly 245 days. At 0.01 m/s², this reduces to about 77 days. For a 100-tonne spacecraft to sustain 0.01 m/s², it requires 1,000 N of thrust. At Isp = 5,000 s and 60% efficiency, the required electrical power is approximately 40.9 MW, well beyond any demonstrated space nuclear power capability. Díaz et al. (2013) confirmed that truly fast human Mars missions require power levels in the tens of megawatts, which remains beyond demonstrated capability.

Ballistic Capture

Ballistic capture exploits three-body gravitational dynamics to insert a spacecraft into Mars orbit without a dedicated propulsive burn. By targeting a trajectory that places the spacecraft within Mars's gravitational sphere of influence along a path that is energetically bound to Mars in the Sun–Mars three-body system, capture can be achieved passively. Luo et al. (2021) demonstrated that combining ballistic capture with aerobraking reduces propellant expenditure and improves insertion flexibility. Mingotti et al. (2011) showed that Earth–Mars transfers combining ballistic escape and low-thrust capture are feasible with reasonable flight times, though transit times typically extend well beyond the Hohmann baseline. Research by Aguiar (2017) found that small spacecraft using this approach require only modest propellant at the cost of transit times of 3.5 years or more, making ballistic capture most relevant for robotic cargo missions where propellant economy vastly outweighs schedule.

6. Comparative Analysis: Efficiency, Duration, and Flexibility

Propellant Efficiency

The propellant efficiency comparison is unambiguous. Consider a spacecraft of 50 tonnes total mass. With chemical propulsion at Isp = 450 s (ve ≈ 4,415 m/s) and a total delta-v of 3.6 km/s, the rocket equation gives:

$$m_0/mf = e^{(\Delta v/ve)} = e^{(3600/4415)} \approx 2.26$$

This implies that approximately 44% of departure mass is propellant, roughly 22 tonnes for a 50-tonne spacecraft. At Isp = 5,000 s (ve ≈ 49,050 m/s), the same delta-v gives:

$$m_0/mf = e^{(3600/49050)} \approx 1.075$$

Only 7% of departure mass is propellant, roughly 3.5 tonnes, increasing the useful mass fraction from 56% to 93%. For cargo missions, this transformation of the propellant budget is mission-enabling. The economic case for MPD propulsion on freight transfers is strong even at modest transit time penalties.

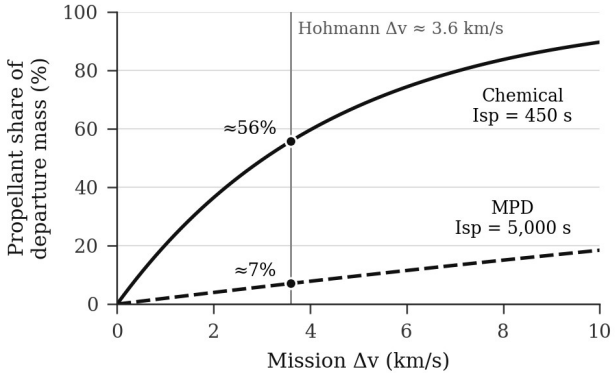


Fig. 8. Propellant share of departure mass from the Tsiolkovsky rocket equation: chemical (Isp = 450 s) vs. MPD (Isp = 5,000 s).

Transit Time

In contrast to this decisive efficiency advantage, the transit time comparison is more constrained. The 259-day Hohmann baseline competes against MPD trajectories that, at near-term power levels of 100 to 500 kW, yield transit times of 200 to 270 days, a reduction of at most 20%. Braun et al. (2021) concluded in a National Academies assessment that while both nuclear thermal and nuclear electric propulsion have the potential to substantially reduce trip time, more research is needed to bring these technologies to operational readiness. The Oleson et al. (2020) opposition-class mission study, targeting a two-year round trip with a 1.5 MWe NEP system, required power levels that pushed the physical limits of near-term reactor technology even to achieve these modest objectives.

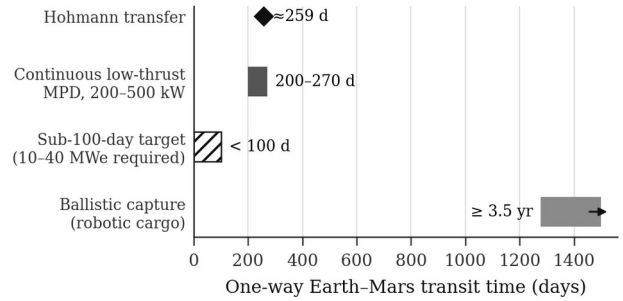


Fig. 9. One-way Earth–Mars transit times for the trajectory options discussed in this paper.

Mission Flexibility

Continuous-thrust trajectories are considerably less constrained by launch windows than Hohmann transfers. Because the spacecraft shapes its trajectory dynamically throughout the mission, a broader range of departure dates is accessible at acceptable propellant cost. For sustained exploration programmes, rather than launching only every 26 months, a continuous-thrust fleet could potentially support launches on a more frequent cadence, accepting modest transit time penalties in exchange for departure flexibility. This represents a qualitative improvement in operational architecture that efficiency and transit time metrics alone do not capture.

Evaluation of Source Reliability

A rigorous assessment of this topic requires acknowledging the limitations of the sources drawn upon, since the quality of any argument depends on the reliability of the evidence underpinning it. Several concerns are worth identifying directly. Sovey and Mantenieks (1991) and Frisbee et al. (1993) are foundational references in MPD and NEP mission design respectively, but both are over three decades old. Electrode materials, power conversion efficiency, and reactor miniaturisation have all advanced substantially since these papers were published, meaning their quantitative performance estimates may be conservative relative to current technology. Where possible, more recent studies (Aftab et al., 2024; Zheng et al., 2021; Mason et al., 2022) have been used to update these figures, but the older works remain the only sources that directly evaluate complete MPD mission architectures at the system level.

The Díaz et al. (2013) paper on VASIMR propulsion is produced by Ad Astra Rocket Company, the commercial entity that manufactures and promotes VASIMR as a competitor to MPD. The paper's optimistic power and transit time projections should therefore be treated with some caution, as the authors have a commercial interest in presenting high-power electric propulsion in the most favourable light. This does not invalidate the physics presented, but it warrants cross-referencing against independent assessments such as Braun et al. (2021), which reaches more conservative conclusions on transit time feasibility.

The radiation dose estimates in Cucinotta et al. (2013) and Guo et al. (2021) carry significant uncertainty ranges. Cucinotta et al. explicitly note that their 95% confidence intervals for radiation-induced mortality span a factor of three to four, reflecting fundamental gaps in understanding of GCR biological effects at the cellular level. The central estimates used in this essay are consistent with the peer-reviewed literature, but they should not be treated as precise predictions. The broader point, that transit time reduction is clinically meaningful, holds across the full uncertainty range. Finally, several trajectory optimisation papers (Morante et al., 2021; Wall and Conway, 2005) present computational solutions based on simplified dynamical models that assume two-body heliocentric motion. Real missions involve gravitational perturbations from Jupiter and other bodies, solar wind variation, and propulsion system degradation over time, none of which are fully captured in these models. The transit time figures cited in this essay are therefore best understood as theoretical lower bounds rather than operationally guaranteed values.

7. Human and Mission Factors

Radiation Exposure and Transit Time

The most significant human factor in Mars mission design is galactic cosmic ray (GCR) exposure. GCRs are high-energy protons and heavy ions that penetrate most practical shielding materials, and their biological effects are among NASA's highest-priority research concerns for deep space missions. Cucinotta et al. (2013) estimated that GCR exposure over a Mars mission could result in 5–10% central estimates of radiation-induced mortality, with upper 95% confidence intervals approaching 20%, substantially exceeding NASA's 3% REID limit. McKenna-Lawlor et al. (2012) confirmed that the cruise phase presents the dominant radiation challenge due to cumulative isotropic GCR exposure, and that a large solar energetic particle event during transit could be catastrophic without adequate shielding.

Reduced transit time therefore has direct clinical implications. Guo et al. (2021) estimated that the accumulated GCR dose equivalent for a Hohmann transfer is approximately 0.65 ± 0.24 sievert during solar maximum and 1.59 ± 0.12 sievert during solar minimum, while a fast transfer enabled by advanced nuclear propulsion could reduce these values to approximately 0.2 and 0.5 sievert respectively. Parihar et al. (2016) demonstrated that cosmic radiation exposure produces persistent cognitive dysfunction in animal models, with hippocampal and cortical performance decrements measured 12 and 24 weeks after irradiation. Patel et al. (2020) classified space radiation health effects as a "red" risk for NASA's Human Research Program, the highest priority tier. This evidence establishes that reducing transit time is not merely a scheduling preference. It is a clinical imperative.

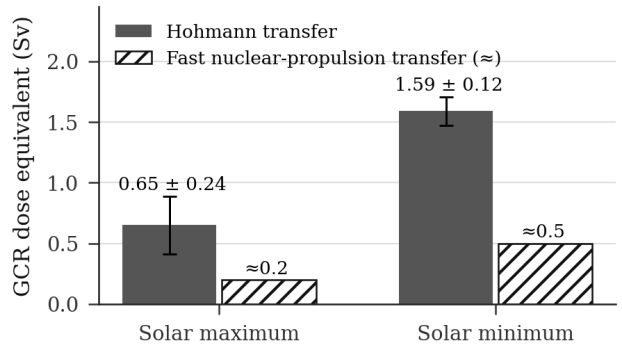


Fig. 10. GCR dose equivalent for a Hohmann transfer vs. a fast nuclear-propulsion transfer (Guo et al., 2021).

Artificial Gravity via Continuous Acceleration

Sustained acceleration can simulate gravity through Newton's second law. The apparent gravitational acceleration experienced by crew aboard a continuously thrusting spacecraft is simply:

$$a = F / m$$

A 100-tonne spacecraft sustaining 0.1 g (0.981 m/s²) would require 98,100 N of thrust. At Isp = 5,000 s and an efficiency of 60%, this demands approximately 4.0 GW of electrical power, roughly four times the output of a large terrestrial nuclear power station. Even at 0.01 g, the power requirement of 400 MW exceeds any space reactor concept currently under serious development. While continuous acceleration-induced artificial gravity is theoretically coherent and would significantly benefit crew health by mitigating bone density loss, muscle atrophy, and fluid redistribution, the power required for clinically meaningful acceleration levels is not achievable with any foreseeable NEP architecture. At realistic MPD thrust levels achievable within the 1 to 5 MWe power range, spacecraft accelerations are on the order of 10^{-4} to 10^{-3} m/s², effectively negligible as a gravity substitute.

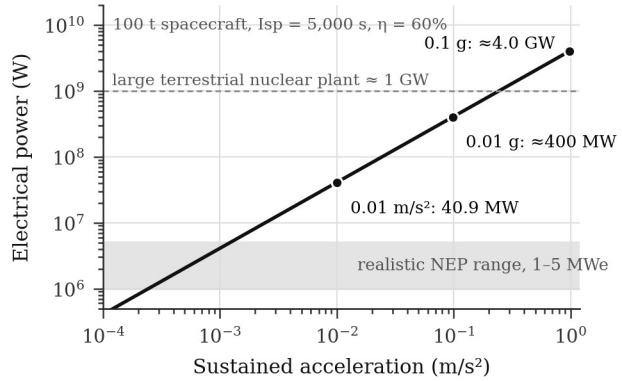


Fig. 11. Electrical power needed to sustain a given acceleration (100 t spacecraft, Isp = 5,000 s, 60% efficiency).

Time Dilation (Relativistic Analysis)

A brief treatment of special relativistic time dilation is warranted, as it is occasionally invoked in popular discussions of fast interplanetary travel. The Lorentz time dilation factor is:

$$t' = t \cdot \sqrt{(1 - v^2/c^2)}$$

At the peak velocity of a brachistochrone Mars trajectory at 0.01 m/s² acceleration, the spacecraft reaches approximately 15 km/s at mid-journey. As a fraction of c (3×10^8 m/s), this is $v/c \approx 5 \times 10^{-5}$. Substituting:

$$t'/t = \sqrt{(1 - (5\times10^{-5})^2)} \approx 1 - 1.25\times10^{-9}$$

The time dilation across a 90-day transit is therefore approximately $90 \times 24 \times 3{,}600 \times 1.25 \times 10^{-9} \approx 0.0097$ seconds, roughly ten milliseconds. This is not merely negligible for mission planning; it is unmeasurable by any biological process. Even in the most optimistic brachistochrone scenario, Mars mission velocities are four to five orders of magnitude below the regime where relativistic effects become operationally significant. Time dilation warrants no further consideration in mission design.

8. Practical Engineering Limitations

The theoretical performance advantages of MPD propulsion are well supported by physics. The engineering limitations that constrain those advantages in practice deserve equally rigorous analysis.

Electrode erosion is the primary lifetime constraint for all MPD thruster types. The high current densities required for efficient plasma acceleration erode cathode and anode materials through sputtering, evaporation, and carbon deposition. Sovey and Mantenieks (1991) noted that extended tests of both pulsed and steady-state devices had demonstrated total impulses two to three orders of magnitude below what would be needed for a cargo mission to Mars, and that life-limiting mechanisms were not yet fully characterised at megawatt-class power levels. Applied-field and superconducting designs mitigate this by reducing current density, but the lifetime of any MPD system for a six-month interplanetary mission at high power has not yet been demonstrated in ground testing at full fidelity.

The vacuum facility requirement for ground testing presents a further challenge. Megawatt-class MPD performance testing requires vacuum pressures below 4×10^{-3} Pa and test facilities capable of handling discharge currents above 10 kA. No facility routinely available for propulsion testing currently meets these specifications for sustained long-duration tests at this power level. Performance data informing mission design studies is therefore largely derived from lower-power tests or short-duration pulses, with extrapolation introducing significant uncertainties.

Propellant storage raises practical concerns. Lithium, widely considered the optimal MPD propellant for its Isp advantages, is chemically reactive with water and oxygen. On-orbit handling and storage require dedicated precautions, increasing system complexity and mission risk. Argon is easier to handle but requires cryogenic storage for propellant-efficient tankage and yields lower Isp.

Power processing units (PPUs) represent additional mass and efficiency losses. At the power levels required for high-performance MPD operation, PPUs must condition raw reactor electrical output to the precise voltage and current waveforms required by the thruster. Even at 95% efficiency, a 1 MWe system dissipates 50 kW in the PPU, adding to the waste heat load. Reynolds et al. (2022) found that PMAD (Power Management and Distribution) mass is a significant

and sensitive component of the overall NEP vehicle mass budget for megawatt-class missions.

Finally, the interaction between the thruster plume and the spacecraft presents design constraints often underweighted in trajectory studies. MPD plumes are dense, partially ionised, and carry charged particles that can contaminate optical sensors, deposit conductive films on solar arrays, and generate electromagnetic interference with communication and navigation systems. Spacecraft configuration must account for plume geometry through careful thruster placement and structural design, adding mass and complexity.

9. Overall Evaluation

The evidence presented across the preceding sections supports a conclusion that is clear in direction but conditional in magnitude. MPD thrusters enable more efficient Earth–Mars transfers than the Hohmann orbit under virtually any realistic power assumption, and they enable faster transfers only under power assumptions that currently remain aspirational.

The propellant efficiency advantage is not conditional. It follows directly from the Isp differential and the structure of the rocket equation. At Isp = 5,000 s versus 450 s, propellant mass requirements for equivalent delta-v fall by roughly an order of magnitude. The NASA Mars Transportation Assessment Study concluded that NEP with MPD or similar high-Isp thrusters is a viable and attractive architecture for both cargo and crew missions, provided that reactor and thermal management technology advances to the required level (Mason et al., 2022).

The transit time advantage is power-limited and currently modest. At 200 to 500 kW, achievable transit times of 200 to 240 days represent a 10 to 20% improvement over the 259-day Hohmann baseline. This is insufficient to materially reduce crew radiation exposure, which scales directly with time in deep space. To achieve sub-100-day transits that would bring GCR dose within acceptable limits, power levels of 10 to 40 MWe are required for spacecraft of reasonable mass.

The trajectory design space enabled by continuous thrust is genuinely richer than the two-impulse Hohmann framework. Continuous heliocentric transfers, spiral escapes, brachistochrone profiles, and ballistic capture all represent valid mission architectures with specific advantages in particular mission contexts. However, the propulsion and power systems required to exploit this design space at mission-relevant thrust levels are not yet mature. The Hohmann transfer, for all its limitations, offers one attribute that no MPD mission currently matches: it can be executed with proven technology today, at known cost, with well-characterised risk.

The most significant counterargument to MPD propulsion as a near-term solution is not physical but programmatic. The currently funded Mars mission architectures, including NASA's Moon-to-Mars programme and SpaceX's Starship architecture, are built entirely around chemical propulsion and aerocapture. No currently funded mission uses or plans to use NEP or MPD propulsion for the primary interplanetary transit phase. The reasons are not limited to technology readiness. Space-qualified megawatt-class reactors require regulatory approval under international nuclear treaty frameworks, impose political risk associated with launch failures scattering radioactive material, and demand new ground support infrastructure. These barriers are not insurmountable, but they mean that the pathway from demonstrated MPD laboratory performance to an operational crewed Mars mission involves decades of regulatory, political, and industrial work alongside the engineering development. Framing MPD purely as a technical problem, as much of the literature implicitly does, may therefore overstate how quickly its theoretical advantages can be realised in practice. This does not diminish the physical case for MPD propulsion; it does, however, mean that the Hohmann transfer will remain the operative baseline for any mission flown before approximately 2040, regardless of what advanced propulsion concepts are theoretically available.

10. Review and Evaluation

This section provides a reflective evaluation of the project as required by the EPQ assessment framework, addressing the successes and limitations of the research process, what would be approached differently given the opportunity to repeat the project, and what broader lessons can be drawn from the experience of undertaking this investigation.

Successes

The most significant success of this project was the depth and quality of peer-reviewed sourcing achieved through the Consensus academic search engine. Access to 22 journal papers, NASA technical reports, and National Academies assessments allowed the argument to be grounded in primary literature rather than secondary summaries or popular science accounts. The decision to include quantitative analysis throughout, including specific impulse comparisons, rocket equation calculations, and power requirement estimates, significantly strengthened the analytical rigour of the essay and produced concrete, defensible conclusions rather than vague generalisations.

The structure of the argument also worked well. Beginning with the Hohmann transfer as a clearly defined baseline before introducing MPD propulsion as an alternative meant the comparison remained grounded throughout. The decision to evaluate human factors and engineering limitations as separate sections, rather than embedding them within the propulsion or trajectory analysis, prevented the argument from becoming circular and ensured each constraint was addressed on its own terms.

Weaknesses and Limitations

The most significant limitation of this project is the absence of primary data. All quantitative figures are drawn from published simulation studies, laboratory experiments, or mission architecture analyses. None have been independently verified or derived from first principles beyond the level of the equations presented in the essay. Where studies use different assumptions, for example

different spacecraft masses, payload fractions, or thruster configurations, direct numerical comparison between them is imprecise. The transit time figures cited for continuous-thrust trajectories are particularly sensitive to assumed spacecraft mass, which varies considerably across the sources consulted.

A further limitation is the date range of some sources. The Frisbee (1993) and Sovey and Mantenieks (1991) studies remain important references because they are among the very few that evaluate complete MPD system architectures at mission scale, but their quantitative estimates of reactor specific mass and electrode lifetime are based on technology that has since advanced. More recent full-system analyses at megawatt power levels are not widely available in open literature, which reflects the relatively low level of funded research in this area rather than a failure of the literature search. The essay attempted to mitigate this by cross-referencing older system-level studies with newer component-level data from recent thruster and reactor papers, but this substitution is imperfect.

The decision not to include nuclear thermal propulsion (NTP) as a third comparison point is also, in retrospect, a limitation. NTP offers transit time reductions comparable to or exceeding near-term NEP/MPD at significantly lower development risk, and its omission means the essay presents a binary comparison when the actual trade space has at least three major options. Including NTP would have required extending the word count substantially, but even a brief comparative treatment would have strengthened the evaluation of where MPD propulsion sits within the wider propulsion landscape.

What Would Be Done Differently

Given the opportunity to repeat this project, three changes would be made. First, the scope would be expanded to include nuclear thermal propulsion as a third trajectory case, providing a more complete picture of the propulsion trade space and contextualising MPD's position more precisely relative to nearer-term advanced options. Second, an attempt would be made to access NASA's Technical Reports Server directly, which contains internal mission architecture studies not always available through academic search engines. Some of the most detailed NEP vehicle concept work is in grey literature reports that did not appear in the Consensus search results. Third, it would have been valuable to contact an aerospace engineering academic or professional with experience in electric propulsion mission design, to provide an expert perspective on which engineering constraints are genuinely binding versus which may be resolved within realistic development timescales. The Edexcel EPQ framework explicitly supports the use of interviews as a primary source, and this would have added a qualitative dimension absent from the current essay.

Lessons Learned

The most important lesson from this project concerns the gap between theoretical performance and operational capability. In every section of this essay, the physics supported a more optimistic conclusion than the engineering evidence sustained. This pattern, where a propulsion technology is demonstrably superior in the laboratory but faces severe constraints when scaled to mission relevance, appears to be a recurring feature of advanced space propulsion development, not a peculiarity of MPD thrusters. Recognising this distinction early would have sharpened the research questions from the outset and led to a more focused investigation of the specific constraints rather than the theoretical performance envelope.

The project also reinforced the value of academic literature as a primary source of evidence, and the importance of evaluating sources critically rather than accepting quantitative claims at face value. The discovery that Díaz et al. (2013) was produced by a company with commercial interests in the technology being evaluated, or that Cucinotta et al.'s radiation mortality estimates carried uncertainty ranges spanning a factor of four, are both findings that required active critical engagement with the sources rather than passive reading. These habits of source evaluation are directly transferable to undergraduate-level research and beyond.

11. Conclusion

Magnetoplasmadynamic thrusters enable meaningfully more propellant-efficient Earth–Mars trajectories than the Hohmann transfer orbit under essentially all realistic power scenarios. This follows directly from the Isp advantage and is not conditional on any technological advance beyond demonstrated thruster physics. An MPD-propelled spacecraft at Isp = 5,000 s requires approximately one-tenth the propellant mass of a chemical equivalent for the same delta-v, transforming mission mass budgets and enabling either heavier payloads or substantially reduced launch costs.

On transit time, the situation is more constrained. Near-term NEP systems in the 100 to 500 kWe range produce modest improvements over the Hohmann baseline, reducing transit times by 10 to 20%. These improvements are insufficient to materially alter the radiation risk profile of crewed missions. Transit times that would achieve meaningful radiation reduction, below 100 days, require power levels in the tens of megawatts, which remain beyond demonstrated space-power capability. The engineering challenges of reactor mass, thermal rejection, electrode erosion, and propellant management all constrain achievable performance below theoretical limits.

The trajectory design space opened by continuous thrust is qualitatively richer than the Hohmann framework, encompassing brachistochrone profiles, optimised heliocentric transfers, and ballistic capture strategies that offer mission-relevant advantages for specific applications. However, the power infrastructure to exploit the most compelling of these profiles does not yet exist.

The overall judgement must therefore be carefully calibrated. MPD thrusters outperform the Hohmann transfer decisively on efficiency, and modestly on transit time under near-term power assumptions. They could outperform it transformatively on transit time under power assumptions that NASA itself projects as achievable only in the late

2030s. The conditions under which MPD propulsion becomes a paradigm-shifting technology for human Mars missions are not physically implausible; they are engineering milestones. Until those milestones are reached, the Hohmann transfer remains the operational baseline, and MPD propulsion represents a compelling, evidence-grounded, but as yet unrealised improvement upon it.

12. References

Aftab, H., et al. “Exploring Efficiency in Next Generation High Temperature Superconducting-Enhanced Applied Field Magnetoplasmadynamic Thrusters: A Combined Numerical and Experimental Study.” Acta Astronautica, 2024. https://consensus.app/papers/details/90967c80136d5cbf931adf18dad30683/

Aguiar, G. “Earth–Mars Low-Thrust Transfers with Ballistic Capture and Real-Solar-System Dynamics.” MSc thesis, 2017. https://consensus.app/papers/details/b439c1a65f335a7988c1f177b9e83214/

Braun, Bobby, et al. Space Nuclear Propulsion for Human Mars Exploration. National Academies of Sciences, Engineering, and Medicine, 2021. https://consensus.app/papers/details/ac1a0d9f44765d51836744fe250cda71/

Boscolo, D., et al. “Dose Limits and Countermeasures for Mitigating Radiation Risk in Moon and Mars Exploration.” Physics, vol. 4, 2022, pp. 1–15. https://consensus.app/papers/details/ffd0bc1f17d05665bb1f2157b37ca361/

Cucinotta, Francis A., et al. “How Safe Is Safe Enough? Radiation Risk for a Human Mission to Mars.” PLOS ONE, vol. 8, no. 10, 2013, e74988. https://consensus.app/papers/details/30f5cab732165facab2bc28c2681344d/

Díaz, Franklin Chang, et al. “Fast and Robust Human Missions to Mars with Advanced Nuclear Electric Power and VASIMR Propulsion.” Ad Astra Rocket Company, 2013. https://consensus.app/papers/details/bd593ff49f745bc1ae76d1e4268932bb/

Frisbee, Robert H., et al. “SP-100 Nuclear Electric Propulsion for Mars Cargo Missions.” AIAA Paper 93-2092, 1993. https://consensus.app/papers/details/e51038fa0fdb584db40cd0c43b4d3db0/

Guo, Jingnan, et al. “Radiation Environment for Future Human Exploration on the Surface of Mars: The Current Understanding Based on MSL/RAD Dose Measurements.” The Astronomy and Astrophysics Review, vol. 29, 2021, article 8. https://consensus.app/papers/details/17c133edea095b98a83ec78709e27d57/

Luo, Zongfu, et al. “Mars Orbit Insertion via Ballistic Capture and Aerobraking.” Astrodynamics, vol. 5, 2021, pp. 167–181. https://consensus.app/papers/details/2431af0b358b5f9f8bae93ffb95b2ada/

Mason, Lee, et al. “Nuclear Power Concepts and Development Strategies for High-Power Electric Propulsion Missions to Mars.” Nuclear Technology, vol. 208, no. 9, 2022. https://consensus.app/papers/details/546b1ebe51465edd8564f050fd23acba/

McKenna-Lawlor, S., et al. “Overview of Energetic Particle Hazards During Prospective Manned Missions to Mars.” Planetary and Space Science, vol. 63–64, 2012, pp. 123–132. https://consensus.app/papers/details/72f4d7928b77520d8bed076118a6ebcc/

Mingotti, G., et al. “Earth–Mars Transfers with Ballistic Escape and Low-Thrust Capture.” Celestial Mechanics and Dynamical Astronomy, vol. 110, no. 2, 2011, pp. 169–188. https://consensus.app/papers/details/adfeab962ae357de923e932d393c164b/

Morante, David, et al. “A Survey on Low-Thrust Trajectory Optimization Approaches.” Aerospace, vol. 8, no. 3, 2021, article 88. https://consensus.app/papers/details/3fb2754cc62856ea99799b49b7451f55/

Oleson, S., et al. “A Combined Nuclear Electric and Chemical Propulsion Vehicle Concept for Piloted Mars Opposition Class Missions.” AIAA 2020-3888, 2020. https://consensus.app/papers/details/5ddae08ea08f55de86262b5dff885b7e/

Parihar, V., et al. “Cosmic Radiation Exposure and Persistent Cognitive Dysfunction.” Scientific Reports, vol. 6, 2016, article 34774. https://consensus.app/papers/details/7c08f1cbee545382b248994c5e388ffe/

Patel, Zarana S., et al. “Red Risks for a Journey to the Red Planet: The Highest Priority Human Health Risks for a Mission to Mars.” npj Microgravity, vol. 6, 2020, article 33. https://consensus.app/papers/details/ba813fd77a165b23abe8db345f657ecc/

Reynolds, Christopher B., et al. “Power Management and Distribution System Trades for an NEP-Based Human Mars Mission.” ASCEND 2022, AIAA 2022-4228, 2022. https://consensus.app/papers/details/7c96673d593751fe84d7dd221962d463/

Sovey, J.S., and M.A. Mantenieks. “Performance and Lifetime Assessment of Magnetoplasmadynamic Arc Thruster Technology.” Journal of Propulsion and Power, vol. 7, no. 1, 1991, pp. 71–83. https://consensus.app/papers/details/da626ed4e02d5aa88ed0b24ef99fe0c0/

Wall, Bradley J., and Bruce A. Conway. “Near-Optimal Low-Thrust Earth–Mars Trajectories via a Genetic Algorithm.” Journal of Guidance, Control, and Dynamics, vol. 28, no. 5, 2005, pp. 1027–1031. https://consensus.app/papers/details/bd1325092bfa58b3b41f8e5e60dc57f7/

Wu, Peng, et al. “The Performance of a Magnetic Nozzle Enhanced Magnetoplasmadynamic Thruster.” Acta Astronautica, 2024. https://consensus.app/papers/details/141232bee06d5b82bd2b5949d87a20fe/

Yuan, Fei-xiang, et al. “Design and Optimization of Space Nuclear Power System with sCO_2 Brayton Cycle on Mars.” Progress in Nuclear Energy, vol. 168, 2024, article 104995. https://consensus.app/papers/details/6036a6f80db7530f827b24d5cfadd355/

Zheng, Jinxing, et al. “Integrated Study on the Comprehensive Magnetic-Field Configuration Performance in the 150 kW Superconducting Magnetoplasmadynamic Thruster.” Scientific Reports, vol. 11, 2021, article 14439. https://consensus.app/papers/details/e7c7d168b1395ee5b23cb4c5fa2d2231/

Appendix A

A Gallium-Fueled Fusion-Choke Applied-Field MPD Thruster: Conceptual Architecture, Performance Analysis, and Feasibility Assessment for Deep-Space Propulsion

Abstract

Chemical propulsion imposes a fundamental mass-fraction penalty on high-Δv interplanetary missions because of the low exhaust velocities achievable through combustion. This paper presents a conceptual analysis of a hybrid deep-space propulsion architecture that combines an Applied-Field Magnetoplasmadynamic (AF-MPD) thruster with a magneto-inertial fusion (MIF) ignition stage and a capillary-fed liquid-gallium electrode system. In the concept, a magnetically confined plasma jet is compressed through a converging magnetic throat to Z-pinch conditions, briefly igniting a deuterium-tritium fusion reaction that heats a secondary propellant of liquid gallium. The gallium is delivered to refractory-metal electrodes through a porous capillary structure and is ablated by the plasma arc, cooling the electrode surface while supplying the bulk of an electromagnetically accelerated exhaust stream. This paper develops and evaluates the governing physics of the concept, including propellant mass-fraction estimates from the Tsiolkovsky rocket equation, fusion energy requirements, capillary pore sizing from the Young–Laplace relation, and neutron transport considerations for the tungsten electrode structure. The analysis indicates that exhaust velocities on the order of tens of kilometers per second could substantially reduce propellant mass fraction relative to chemical propulsion, consistent with performance ranges reported for other magneto-inertial fusion propulsion concepts in the literature. However, the assessment also identifies several unresolved engineering and programmatic constraints, including capillary pore tolerances near the limits of current additive manufacturing processes, neutron-induced transmutation of the tungsten electrode structure, exhaust-driven contamination of spacecraft surfaces, constrained global tritium supply, liquid-metal slosh dynamics, and regulatory ambiguity under existing space treaty frameworks. Realistic near-to-mid-term mission applications are more plausibly found in uncrewed cargo and science missions than in crewed transit, given the outstanding radiation, contamination, and safety-qualification questions identified. The paper concludes by outlining the ground-based manufacturing, fuel-cycle, and regulatory infrastructure required to mature the concept toward flight readiness.

Keywords: Magneto-Inertial Fusion Propulsion; Applied-Field MPD Thruster; Z-Pinch; Liquid-Metal Propellant; Capillary Porous System; Deep-Space Propulsion

Nomenclature

Δv	characteristic velocity increment, m/s
ve	effective exhaust velocity, m/s
mp	propellant mass, kg
md	dry (vehicle) mass, kg
Ek	kinetic energy, J
m, v	mass, kg; velocity, m/s
γ	surface tension, N/m
ΔP	capillary (Laplace) pressure, Pa
r	pore radius, m
ρ	density, kg/m³
n	number density, atoms/m³ (context-dependent: gallium atomic density in Sec. A4.2; total ion density in Secs. A3.5-A3.6)
σ	microscopic neutron cross-section, barns (1 b = 10^{-28} m²)
λ	neutron mean free path, m
NA	Avogadro constant, 6.022×10^{23} mol^{-1}
$g0$	standard gravity, 9.81 m/s²
T	plasma temperature, keV (or K)
V	plasma volume, m³
τE	energy confinement (dwell) time, s
Rf	fusion reaction rate density, m^{-3}s^{-1}
Pf	fusion power, W
Wp	stored plasma thermal energy, J
β	plasma beta, dimensionless
B	magnetic field, T
$\mu0$	vacuum permeability, $4\pi\times10^{-7}$ T·m/A
I	pinch/coil current, A
kB	Boltzmann constant, 1.381×10^{-23} J/K
L	effective current-carrying length (AF-MPD), m
ηF	electromagnetic thrust-coupling factor, dimensionless

Acronyms/Abbreviations

AF-MPD	Applied-Field Magnetoplasmadynamic thruster
MIF	Magneto-Inertial Fusion
CPS	Capillary Porous System
LPBF	Laser Powder Bed Fusion
LME	Liquid Metal Embrittlement
D-T	Deuterium-Tritium
OST	Outer Space Treaty (1967)
PTBT	Partial (Limited) Test Ban Treaty (1963)
COSPAR	Committee on Space Research
ACS	Attitude Control System
NSPM-20	National Security Presidential Memorandum 20

A1. Introduction

Interplanetary and deep-space missions are fundamentally constrained by the propellant mass fraction implied by the Tsiolkovsky rocket equation. For chemical propulsion systems, whose exhaust velocity is limited to approximately 4.4 km/s for high-performance liquid-hydrogen/liquid-oxygen combinations, even modest mission velocity increments (Δv) of several kilometers per second impose propellant mass fractions that dominate vehicle design. This constraint has motivated decades of research into higher-specific-impulse alternatives, including electric propulsion, nuclear thermal propulsion, and a range of fusion-based propulsion concepts that couple pulsed, magnetically confined fusion reactions to electromagnetic or magnetic-nozzle acceleration stages.

This paper examines a conceptual propulsion architecture that combines three previously distinct technology threads: (i) an Applied-Field Magnetoplasmadynamic (AF-MPD)

thruster, which uses an externally applied magnetic field to improve thrust efficiency and electrode life relative to self-field MPD designs; (ii) a magneto-inertial fusion (MIF) ignition stage, in which a magnetically confined plasma column is compressed to fusion-relevant density and temperature through a Z-pinch process; and (iii) a capillary porous system (CPS) liquid-metal electrode, in which a refractory-metal skeleton is continuously replenished with liquid gallium delivered by capillary action. In the concept, plasma injected into a converging magnetic throat undergoes Z-pinch compression sufficient to briefly ignite a deuterium-tritium (D-T) fusion reaction; gallium ablated from the porous electrode by the resulting plasma arc absorbs the associated thermal load while supplying the bulk of the accelerated propellant mass.

The purpose of this paper is to formalize the concept's operating principles, quantify its performance envelope using standard propulsion and plasma-physics relations, and evaluate the manufacturing, materials, operational, and regulatory constraints implied by the design. Physics and engineering calculations are performed where the underlying parameters can be sourced from established reference data; where a value is assumed rather than sourced, it is explicitly flagged as a placeholder requiring further verification, consistent with a zero-fabrication standard for this manuscript. Section A2 describes the system architecture. Section A3 develops a performance analysis. Section A4 addresses manufacturing and materials constraints. Section A5 discusses operational risk factors. Section A6 addresses regulatory and planetary-protection considerations. Section A7 proposes candidate mission profiles. Section A8 outlines supporting ground infrastructure. Section A9 concludes.

A2. System Architecture and Operating Principles

A2.1 Overview

The proposed thruster integrates three functional subsystems into a single electromagnetic acceleration stage. First, a plasma gun injects and accelerates a deuterium-tritium fuel mixture into a magnetic throat of decreasing cross-section, where the applied magnetic field intensifies and drives geometric and magnetic compression of the plasma column. Second, the resulting Z-pinch discharge briefly ignites a fusion reaction that heats and expands the plasma; the associated thermal flux is absorbed at the electrode surface by a continuously replenished, ablating layer of liquid gallium supplied through a porous refractory-metal (tungsten or molybdenum) capillary structure. Third, a radial applied magnetic field traps the resulting energetic electron population, which electromagnetically drags the co-produced gallium ions through a magnetic nozzle to form the exhaust plume. These subsystems are described in turn below.

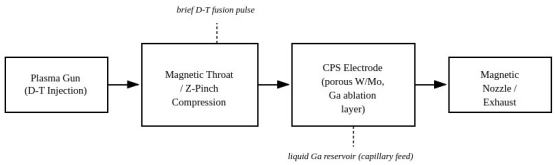


Fig. A1. System architecture: plasma injection, magnetic throat / Z-pinch compression, CPS gallium electrode, and magnetic-nozzle exhaust.

A2.2 Multi-Stage Magnetic Compression and Z-Pinch Ignition

Fuel injection follows conventional plasma-gun principles: a deuterium-tritium (or deuterium–helium-3) mixture is ionized and accelerated electromagnetically to hypersonic velocity before entering the converging magnetic throat. As the plasma column advances into the region of increasing applied field strength, both the physical channel cross-section and the magnetic mirror ratio increase the confining pressure on the plasma. At the throat's narrowest point, the axial discharge current produces a self-pinching $j \times B$ force (the Z-pinch effect) that compresses the plasma column further, in principle to densities and temperatures sufficient to briefly satisfy Lawson-type ignition criteria for a D-T reaction.

The resulting fusion burn is necessarily pulsed rather than steady-state. This is consistent with the operating principle of comparable published concepts: NASA Marshall Space Flight Center's Z-pinch magneto-inertial fusion propulsion design study [1] and MSNW LLC's magneto-inertial Fusion Driven Rocket concept [2] are both organized around repetitively pulsed operation rather than continuous fusion burn. The specific pulse repetition rate assumed for this concept (20 Hz) is not independently derived or sourced in the underlying design notes.

[Flagged placeholder: the 20 Hz pulse-repetition-rate assumption requires supporting analysis of driver electronics, thermal cycling, and structural fatigue limits before it can be treated as a validated design parameter.]

A2.3 Capillary-Fed Liquid-Gallium Electrode System

A conventional solid electrode, whether tungsten, copper, or a refractory alloy, would be rapidly eroded or vaporized by direct exposure to fusion-relevant heat and particle flux. The concept instead adopts a consumable, self-replenishing electrode strategy drawn from capillary porous system (CPS) liquid-metal ion source design: the electrode is manufactured by Laser Powder Bed Fusion (LPBF) additive manufacturing as a porous refractory-metal skeleton, through which liquid gallium is continuously supplied from a propellant reservoir.

Iron-, nickel-, and chromium-based alloys, including Inconel, are unsuitable structural materials for this application because gallium is a well-documented embrittlement agent for such alloys at elevated temperature, penetrating and weakening grain boundaries through liquid metal embrittlement (LME) [3]. Tungsten and molybdenum are comparatively resistant to gallium-induced LME and are therefore the assumed skeleton materials. During operation, the plasma arc vaporizes and erodes the gallium wetting the

pore surface; this ablation is deliberate rather than incidental — it removes thermal energy from the underlying refractory structure through the latent heat of vaporization while simultaneously generating the majority of the thruster's propellant mass flow. This general approach is consistent with demonstrated porous-tungsten and capillary liquid-metal ion source designs used in electric propulsion and ion-beam applications [4].

A2.4 Plasma Acceleration and Charge Configuration

Two charge-configuration strategies were considered in the underlying concept notes. Biasing the outer shell strongly positive to repel gallium ions and preferentially accelerate electrons was rejected because it is subject to the space-charge limit: a dense electron sheath forms at the wall, both shielding the applied field and exposing the structure to destructive electron bombardment. The adopted configuration instead follows conventional Hall- and MPD-thruster practice [5], in which a radial applied magnetic field traps energetic electrons in an azimuthal drift layer. This electron population both sustains ionization of the gallium sweating from the electrode surface and establishes the axial electric field that electromagnetically accelerates the resulting gallium ions rearward through a magnetic nozzle, producing a collimated exhaust plume.

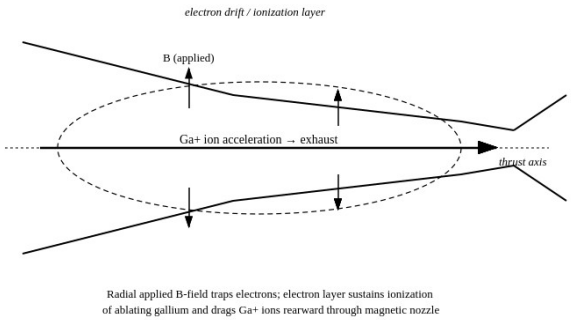


Fig. A2. Radial applied-field charge configuration: electron drift layer sustains ionization and drags Ga+ ions through the magnetic nozzle.

A3. Performance Analysis

A3.1 Exhaust Velocity Regime and Comparison to Published Concepts

The concept notes propose an electromagnetic exhaust velocity range of 20–100 km/s, driven by the combination of heavy gallium ions (atomic mass ≈ 69.7 u) with fusion-augmented acceleration. This range corresponds to a specific impulse (Isp = ve/g0) of approximately 2,040–10,190 s. For comparison, NASA Marshall Space Flight Center's Z-pinch magneto-inertial fusion propulsion concept design study reports an Isp near 19,436 s at a thrust level of several kN [1], while MSNW LLC's magneto-inertial Fusion Driven Rocket concept reports an Isp range of approximately 1,606–5,722 s depending on assumed fusion gain, with burn durations of one to ninety days [2].

The gallium-fusion concept's assumed performance range therefore falls within, or close to, the range reported for comparable published magneto-inertial fusion propulsion studies, which provides qualitative plausibility for the assumed exhaust velocity but does not by itself validate the specific engine design proposed here. A first-principles derivation of achievable exhaust velocity from the concept's magnetic field topology, pinch current, and gallium ionization fraction remains outstanding and is recommended as follow-on work. For the calculations in Sections A3.2–A3.3, a representative exhaust velocity of ve = 50 km/s (Isp ≈ 5,097 s) is adopted as a mid-range design point, consistent with the value used in the underlying concept notes.

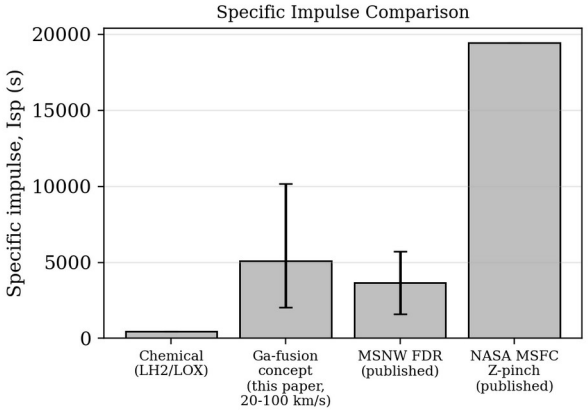


Fig. A3. Specific impulse of this concept (range) versus chemical propulsion and published MIF concepts (MSNW FDR; NASA MSFC Z-pinch).

A3.2 Propellant Mass-Fraction Estimate

The Tsiolkovsky rocket equation relates the propellant mass mp required to achieve a velocity increment Δv, given dry vehicle mass md and effective exhaust velocity ve:

$$mp = md \times (e^{(\Delta v / ve)} - 1) \quad (1)$$

For an illustrative dry mass of md = 100 t and a representative impulsive trans-Mars-injection Δv of 8 km/s, a chemical propulsion system with ve = 4.4 km/s requires an exponent Δv/ve = 1.82, giving mp ≈ 5.16 md ≈ 516 t. Under the same assumptions, the gallium-fusion concept's ve = 50 km/s design point gives an exponent of 0.16 and mp ≈ 0.17 md ≈ 17 t — an approximately 30-fold reduction in propellant mass for the same mission Δv. Both figures were independently recomputed for this paper and are consistent with the underlying concept notes.

This comparison carries an important caveat that is not addressed in the underlying concept notes. The Tsiolkovsky relation in its impulsive form is appropriate for short, high-thrust burns, as is characteristic of chemical propulsion. Magneto-inertial fusion propulsion concepts of the type considered here instead produce thrust at milli-g-class accelerations; NASA Marshall Space Flight Center's Z-pinch propulsion trajectory analysis explicitly treats this regime as “medium thrust” and requires numerically integrated, patched-conic trajectory analysis rather than an impulsive Δv budget [1]. A low-thrust vehicle typically requires a larger effective Δv than an equivalent impulsive trajectory, because of gravity losses accrued during a slow escape spiral out of a planetary gravity well. The 8 km/s, 17 t propellant estimate above should therefore be read as an optimistic lower bound; a credible mission-level propellant

budget for this concept requires low-thrust trajectory optimization rather than the impulsive rocket equation alone.

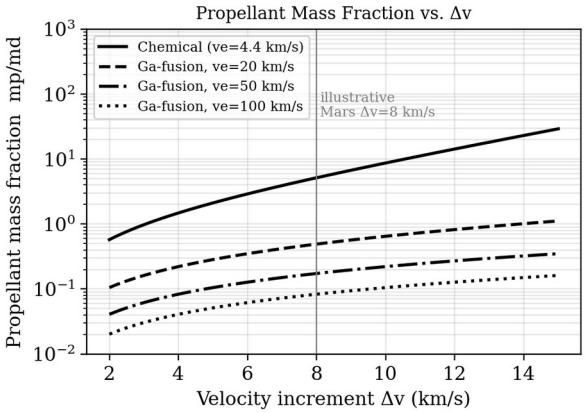


Fig. A4. Propellant mass fraction vs. Δv (Tsiolkovsky equation) for chemical propulsion and this concept's exhaust-velocity range.

A3.3 Fusion Fuel Mass per Pulse

An independent scenario in the concept notes considers the D-T fuel mass required to accelerate a 65 t gallium propellant load to the design exhaust velocity of 50 km/s. The kinetic energy required is:

$$Ek = \tfrac{1}{2}\, m\, v^2 = \tfrac{1}{2}(65{,}000\ \text{kg})(50{,}000\ \text{m/s})^2 \approx 8.1\times10^{13}\ \text{J} \quad (2)$$

D-T fusion releases 17.6 MeV per reaction. Given the combined mass of one deuteron and one triton (≈ 5.03 u ≈ 8.35×10^{-27} kg), this corresponds to a fusion energy density of approximately 3.4×10^{14} J per kilogram of D-T fuel consumed, a figure independently recomputed for this paper from standard nuclear reaction data and consistent with the concept notes. Dividing the required kinetic energy by this energy density gives a nominal D-T fuel mass of:

$$mDT = Ek / (3.4\times10^{14}\ \text{J/kg}) \approx 0.24\ \text{kg} \quad (3)$$

This calculation, however, implicitly assumes that 100% of the fusion yield is converted into propellant kinetic energy, with no losses to radiation, imperfect magnetic confinement, incomplete burn fraction, or inefficient momentum coupling between the fusion plasma and the gallium propellant. Conceptual Z-pinch propulsion studies in the literature explicitly model these coupling losses and report substantially lower net propulsive efficiencies than an idealized 100% conversion would imply [6].

[Flagged placeholder: the fusion-to-propellant energy coupling efficiency assumed for this concept is not stated in the underlying design notes — Insert Data Here. The 0.24 kg D-T fuel-mass figure should accordingly be treated as a strict lower bound rather than an engineering estimate.]

[Flagged conflict: the 65 t gallium propellant load used in this scenario is not reconciled with the 17 t propellant-mass-fraction result obtained in Section A3.2, which was derived under a separate set of assumptions (100 t dry mass, 8 km/s impulsive Δv). These two figures originate from independent illustrative scenarios in the underlying concept notes rather than a single, self-consistent vehicle mass budget, and should be reconciled before being combined into an overall vehicle sizing estimate.]

A3.4 Staged Energy and Power Balance

Section A3.3 computed the D-T fuel mass required to accelerate the 65 t gallium propellant load to ve = 50 km/s under an idealized 100% fusion-to-propellant energy conversion. This section replaces that idealization with a staged energy budget and an explicit, labeled coupling-efficiency parameter, per the zero-fabrication standard used throughout this manuscript.

Define the overall fusion-to-propellant coupling efficiency:

$$\eta_{total} = E_{useful} / E_{fusion} \quad (6)$$

where E_useful is the useful exhaust kinetic energy delivered to the propellant (taken as Ek from Eq. 2, 8.1×10^{13} J, for the same 65 t / 50 km/s baseline scenario used in Section A3.3) and E_fusion is the total D-T fusion energy that must be released to deliver it. Between fusion energy release and useful exhaust kinetic energy, energy is diverted through several loss channels: (i) gallium heating, vaporization, and ionization at the ablating electrode surface; (ii) incomplete D-T burn fraction (quantified directly in Section A3.5); (iii) the 14.1 MeV neutron carrying 80.1% of the 17.6 MeV D-T yield, most of which escapes the thin gallium ablation layer without depositing energy, consistent with the ~9-10 cm neutron mean free path estimated in Section A4.2; (iv) radiation (bremsstrahlung, line radiation) and conductive thermal losses from the pinch plasma; (v) resistive/Ohmic and momentum-coupling losses in the electromagnetic acceleration stage (Section A2.4); and (vi) magnetic-nozzle divergence and incomplete plasma detachment.

Independent, validated values for each of these six channels are not established in the underlying concept notes or in the literature cited for this specific engine architecture. Consistent with the zero-fabrication standard, this paper does not multiply out a bottom-up product of individually fabricated sub-efficiencies; instead, η_total is treated directly as a swept design parameter over η_total = 5%, 10%, 20%, 30%, with 10% adopted as the nominal engineering assumption. This nominal value is broadly consistent with the substantially-lower-than-ideal net propulsive efficiencies reported for comparable pulsed Z-pinch propulsion coupling analyses in the literature [6], but is not independently derived for this architecture.

[Flagged placeholder: per-channel efficiency values (heating/vaporization/ionization loss, burn fraction, neutron loss, radiation loss, EM coupling loss, nozzle loss) are not independently quantified in this paper; only their aggregate product, η_total, is treated as an assumed sensitivity-swept parameter — Insert Data Here: per-channel coupling-efficiency breakdown from dedicated simulation.]

Table A1 also requires a total-mission burn duration to convert the total energy/fuel-mass figures below into average fusion power and per-pulse energy at the 20 Hz repetition rate assumed in Section A2.2. This duration is not stated in the underlying concept notes; a 1-90 day bracket is adopted here by analogy to the burn-duration range reported for MSNW's comparable magneto-inertial Fusion Driven

Rocket concept (Section A3.1) [2], rather than as a value derived from this concept's own design notes.

Table A1. Energy and power balance vs. η_total (baseline: 65 t Ga, ve = 50 km/s, Eq. 2 gives Euseful = Ek = 8.125×10¹³ J; 20 Hz pulse rate; burn-duration bracket 1-90 days [2]; nominal case in bold).

η_total	E_fusion (J)	m_DT (kg)	P_fusion,avg (W)
5%	1.63×10^{15}	4.78	1.9×10^{10}-2.1×10^{8}
10% (nom.)	8.13×10^{14}	2.39	9.4×10^{9}-1.1×10^{8}
20%	4.06×10^{14}	1.19	4.7×10^{9}-5.2×10^{7}
30%	2.71×10^{14}	0.80	3.1×10^{9}-3.5×10^{7}

P_fusion,avg spans the 1-90 day burn-duration bracket; the corresponding energy per pulse at 20 Hz is P_fusion,avg / 20 in each case (e.g., nominal case: 4.70×10^{8} J/pulse at the 1-day bound, 5.22×10^{6} J/pulse at the 90-day bound). The useful (post-efficiency) exhaust power is, by definition, independent of η_total: Puseful,avg = Euseful / t_burn = 9.40×10^{8} W (1-day burn) to 1.04×10^{7} W (90-day burn), with a corresponding useful energy per pulse of 4.70×10^{7} J to 5.22×10^{5} J. The required fusion power and D-T fuel mass instead scale as 1/η_total. The nominal (η_total = 10%) fusion-power bracket, 1.05×10^{8} - 9.40×10^{9} W, is carried forward as the required fusion power for the ignition check in Section A3.5.

A3.5 Fusion Ignition and Lawson Criterion Analysis

This section tests whether a D-T plasma consistent with the pulsed Z-pinch compression described in Section A2.2 could plausibly produce the required fusion power identified in Section A3.4, without assuming ignition. The underlying concept notes do not specify plasma temperature, density, volume, or confinement time; the following values are adopted as order-of-magnitude assumptions by analogy to the Z-pinch and magneto-inertial fusion (MIF) concepts already cited in this paper [1,2,6], and are flagged accordingly rather than treated as validated design values: plasma temperature T ≈ 10 keV; total ion (deuteron + triton) density $n \approx 1\times10^{25}$ m^{-3}, with an approximately equal (50/50) D/T mixture so that $n_D = n_T = n/2$; a compressed pinch-column volume $V \approx 4\times10^{-6}$ m³ (illustrative cylindrical geometry, radius ≈ 0.5 cm, length ≈ 5 cm); and a confinement (compression dwell) time $\tau_E \approx 1$ μs, representative of pulsed Z-pinch/MIF stagnation timescales rather than steady-state magnetic confinement.

[Flagged placeholder: T, n, V, and τ_E are order-of-magnitude assumptions by analogy to cited MIF/Z-pinch literature, not values derived from this concept's own design notes or from a device-specific simulation — Insert Data Here: verified plasma parameters from dedicated Z-pinch compression modeling.]

For an approximately equal D/T mixture, the fusion reaction rate density is:

$$R_f = n_D \cdot n_T \cdot \langle\sigma v\rangle = (n^2/4)\langle\sigma v\rangle \quad (7)$$

Using the standard D-T reactivity $\langle\sigma v\rangle \approx 1.1\times10^{-22}$ m³/s at T = 10 keV: $R_f = (1\times10^{25})^2/4 \times 1.1\times10^{-22} \approx 2.75\times10^{27}$ reactions·m^{-3}·s^{-1}. The corresponding fusion power is:

$$P_f = R_f \cdot V \cdot E_{DT} \approx (2.75\times10^{27})(4\times10^{-6}\ \text{m}^3)(2.82\times10^{-12}\ \text{J}) \approx 3.05\times10^{10}\ \text{W (30.5 GW)} \quad (8)$$

This instantaneous fusion power (30.5 GW) is of the same order of magnitude as, and exceeds, the nominal required fusion power bracket from Section A3.4 (1.05×10^{8} - 9.40×10^{9} W), indicating no fundamental power-magnitude mismatch at this assumed n, T, V operating point. This comparison alone, however, only tests whether the assumed reacting state — if achieved — could supply enough power; it does not test whether that state is itself achievable, which the Lawson analysis below addresses directly.

The Lawson triple product for the assumed state is:

$$n\cdot T\cdot\tau_E = (1\times10^{25}\ \text{m}^{-3})(10\ \text{keV})(1\times10^{-6}\ \text{s}) \approx 1\times10^{20}\ \text{keV}\cdot\text{s}\cdot\text{m}^{-3} \quad (9)$$

The standard D-T ignition-relevant triple-product criterion is commonly cited in the range $nT\tau_E \gtrsim 3\times10^{21}$ keV·s·m^{-3} (order-of-magnitude; reported values vary by convention and required gain), roughly 30× above the value calculated here. The simpler Lawson breakeven criterion ($n\tau_E \gtrsim \sim1.5\times10^{20}$ s·m^{-3} near optimal temperature) is likewise not met: $n\tau_E = (1\times10^{25})(1\times10^{-6}) = 1\times10^{19}$ s·m^{-3}, about an order of magnitude below breakeven. Under these assumptions, the plasma state does NOT satisfy either the Lawson breakeven or the ignition triple-product criterion; this result is reported directly rather than adjusted to force a positive outcome.

The approximate stored plasma thermal energy is:

$$W_p \approx 3nkBT\cdot V \approx 3(1\times10^{25})(1.60\times10^{-15}\ \text{J})(4\times10^{-6}\ \text{m}^3) \approx 1.89\times10^{5}\ \text{J} \quad (10)$$

Relating this to confinement/loss power through $\tau_E = W_p/P_{loss}$ and inverting for the assumed $\tau_E \approx 1$ μs gives an implied loss power $P_{loss} \approx W_p/\tau_E \approx 1.89\times10^{11}$ W (189 GW) — roughly six times larger than the fusion power P_f calculated above. This is internally consistent with the sub-ignition finding: a plasma approaching ignition would have alpha-heating power comparable to or exceeding confinement losses, whereas here total fusion power (let alone the smaller alpha-only fraction of it) is well below the power needed to sustain the assumed state for τE, confirming this operating point is confinement-loss-dominated rather than self-heating-dominated.

Of the 17.6 MeV D-T yield, the 3.5 MeV alpha particle carries 19.9% (≈6.06 GW of Pf) and the 14.1 MeV neutron carries 80.1% (≈24.4 GW of Pf). Consistent with Section A4.2's neutron mean-free-path estimate, most neutron energy is expected to escape the compressed plasma and thin gallium layer rather than depositing locally. The fraction of alpha energy actually deposited within the assumed centimeter-scale, microsecond-duration pinch volume is not established in this paper or in the literature reviewed.

[Flagged placeholder: alpha energy deposition fraction within the assumed plasma volume and confinement time requires simulation/experimental validation and is not estimated here.]

Distinguishing breakeven, ignition, and engineering feasibility: (1) Breakeven/ignition, in the strict Lawson sense of a self-sustaining, alpha-heating-dominated plasma, is NOT satisfied by the assumed parameters — both the

triple product and the simpler nτE criterion fall short by roughly one to two orders of magnitude, and the implied confinement-loss power exceeds the fusion power produced. (2) This does not by itself disqualify the concept as described, because Section A2.2 already characterizes the design as an externally driven, pulsed Z-pinch rather than a self-igniting steady-state device: the pulsed compression driver, not alpha self-heating, is what establishes the assumed n, T state each cycle, and the D-T yield is properly understood as a supplementary, not dominant, energy contribution relative to that external driver. (3) Engineering feasibility remains undetermined: whether the assumed n, T, V, τE operating point is itself physically achievable by the described magnetic-throat/Z-pinch compression mechanism, and at what driver energy cost, is not established by this paper and requires dedicated compression-dynamics simulation.

A3.6 Plasma Confinement and Magnetic Pressure Requirements

Using the plasma n and T assumed in Section A3.5, and adopting the total (electron + ion) particle density convention for a fully ionized, quasineutral D-T plasma (ne = ni = n for singly-charged D and T ions, so total particle density = 2n), the plasma pressure is:

$$P_{plasma} = (n_e + n_i)kBT = 2n \cdot kBT \approx 2(1\times10^{25})(1.60\times10^{-15}\ \text{J}) \approx 3.20\times10^{10}\ \text{Pa (32.0 GPa)} \quad (11)$$

The confining magnetic field required to balance this pressure at a given plasma beta β = Pplasma/PB is:

$$B_{required} = \sqrt{(2\mu_0 P_{plasma} / \beta)} \quad (12)$$

Adopting β ≈ 0.5 as a representative high-beta operating point (flagged assumption, consistent with the high-beta Z-pinch/FRC-like operation typical of the MIF concepts cited in this paper [1,2,6]) gives Brequired ≈ 401 T. The underlying concept notes describe an "applied magnetic field" qualitatively (Sections A2.2-A2.4) but do not state a numeric design field, so this figure cannot be checked against a stated value and is flagged as requiring simulation/experimental validation against an actual coil/pinch-driver design. Quantitatively, ~400 T is far beyond the ~10-20 T practical ceiling for continuously energized superconducting magnets, but is of the same order as pulsed, self-generated Z-pinch fields at the multi-megaampere current level computed below — consistent with the pulsed, rather than continuously applied, nature of the compression field already described in Section A2.2.

[Flagged placeholder: β ≈ 0.5 is an assumed operating point, not a value stated in the underlying concept notes or independently derived for this architecture — Insert Data Here: verified plasma beta from device-specific simulation.]

The corresponding magnetic energy stored in the pinch volume is:

$$U_B = (B^2/2\mu_0)\cdot V \approx 2.52\times10^5\ \text{J (252 kJ)} \quad (13)$$

For the pinch-column (Z-pinch) geometry already described in Section A2.2, the appropriate field relation is the cylindrical-current Ampere's-law result for the azimuthal self-field of an axial pinch current — not a solenoid or open straight-wire-in-vacuum model — evaluated at the pinch radius r ≈ 0.5 cm assumed in Section A3.5:

$$B_{required} = \mu_0 I / (2\pi r) \implies I \approx 1.00\times10^7\ \text{A (10.0 MA)} \quad (14)$$

A required pinch current of order 10 MA is broadly consistent, in order of magnitude, with the multi-to-tens-of-megaampere currents used in existing pulsed-power Z-pinch research facilities, which supports the plausibility of the current magnitude in isolation. Sustaining a discharge of this magnitude repetitively at the 20 Hz rate assumed in Section A2.2 (rather than as a single-shot pulse, as in existing Z-pinch facilities) is a substantially different and currently unresolved pulsed-power engineering regime.

The associated magnetic-pressure structural stress on any confining structure is:

$$\sigma_B \sim B^2/2\mu_0 = P_{plasma}/\beta \approx 6.41\times10^{10}\ \text{Pa (64.1 GPa)} \quad (15)$$

This is roughly 60-200× the ultimate tensile strength of tungsten or molybdenum alloys (typically a few hundred MPa to ~1 GPa), the candidate electrode structural materials identified in Section A2.3/A4.1. A static structure could not continuously withstand this magnetic pressure. Because the concept is inherently pulsed rather than steady-state — in the manner of existing Z-pinch experiments, where confinement during the brief current pulse is provided by plasma/liner inertia rather than a static containment vessel — the governing engineering question is dynamic/fatigue loading over a sub-microsecond-to-microsecond pulse rather than static stress. This reframing does not resolve the constraint; it remains unquantified and is flagged as requiring dedicated pulsed-structural analysis.

[Flagged placeholder: dynamic/fatigue structural response of the electrode and coil structure under repetitive ~64 GPa-equivalent pulsed magnetic loading at 20 Hz requires simulation/experimental validation and is not assessed in this paper.]

Master Parameters (Sections A3.4-A3.6)

The following table consolidates the parameters used across the energy-balance, Lawson, and magnetic-confinement analyses above, per the consistency rules for this revision. Where the paper's underlying notes contain two independently sourced values for the same quantity (the 65 t and 17 t propellant-mass scenarios), both are retained and their distinct purposes are labeled explicitly rather than one being arbitrarily discarded, consistent with the flagged conflict already noted in Section A3.3.

Parameter (basis/status)	Value
Propellant mass, energy-budget scenario (Sec. A3.3 baseline)	mGa = 65,000 kg
Propellant mass, Δv/mass-fraction scenario (Sec. A3.2 baseline; independent, not reconciled with mGa)	mp ≈ 17,000 kg
Exhaust velocity (Sec. A3.1 design point)	ve = 50 km/s
Pulse repetition rate (Sec. A2.2, flagged, unvalidated)	f = 20 Hz
Total-mission burn duration (assumed, by analogy to ref. [2])	1-90 days
Coupling efficiency (assumed, swept, Sec. A3.4)	ηtotal = 5/10/20/30%, nom. 10%

Parameter (basis/status)	Value
D-T fusion yield (standard data)	EDT = 17.6 MeV
Plasma temperature (assumed, by analogy [1,2,6])	T ≈ 10 keV
Total ion density D+T (assumed, by analogy [1,2,6])	n ≈ 1×10^{25} m^{-3}
D/T mixture (assumed equal mix)	≈50/50 by number
Pinch-column volume (illustrative geometry assumption)	V≈4×10^{-6} m^3 (r≈0.5cm,L≈5cm)
Confinement/dwell time (assumed, by analogy [1,2])	τE ≈ 1 μs
Plasma beta (assumed, high-beta pinch/FRC-like [1,2,6])	β ≈ 0.5
Required confining field (calculated, Sec. A3.6; no paper-stated field to compare against)	Brequired ≈ 401 T
Required pinch current (calculated, Sec. A3.6, Z-pinch geometry)	Irequired ≈ 10.0 MA

A3.7 Physics Feasibility Summary

1. Required fusion power (nominal ηtotal = 10%): ≈1.05×10^{8} - 9.40×10^{9} W average, depending on the assumed 1-90 day burn duration bracket, with ≈5.2×10^{6} - 4.7×10^{8} J per pulse at 20 Hz.

2. Assumed plasma state: T ≈ 10 keV, n ≈ 1×10^{25} m^{-3}, τE ≈ 1 μs — order-of-magnitude assumptions by analogy to cited MIF/Z-pinch literature, not independently derived or simulated for this architecture.

3. Lawson result: nTτE ≈ 1×10^{20} keV·s·m^{-3}, roughly 30× below the ~3×10^{21} keV·s·m^{-3} ignition-relevant triple product; the simpler nτE criterion is also not met. Ignition/breakeven is NOT satisfied under these assumptions; the implied confinement-loss power (≈189 GW) exceeds the calculated fusion power (≈30.5 GW).

4. Required magnetic field: Brequired ≈ 401 T (β ≈ 0.5 assumption) with Irequired ≈ 10.0 MA pinch current; the associated magnetic-pressure structural stress (≈64.1 GPa) is far beyond the static strength of candidate electrode materials, though partially reframed (not resolved) by the concept's inherently pulsed operation.

5. Overall assessment: marginal/uncertain, trending toward inconsistent with the calculated requirements as currently specified. The assumed plasma state does not reach ignition; the required field and structural stress far exceed continuous/static engineering limits; and while the instantaneous fusion power implied by the assumed reacting state is of a plausible order of magnitude relative to the mission energy requirement, this does not establish that the assumed n, T, τE state is achievable by the Z-pinch compression mechanism as described. None of these findings indicate a fundamental physical impossibility, but they substantially raise the bar for the simulation and experimental validation work already recommended in Section A9, and should be read alongside it rather than as a separate conclusion.

A3.8 AF-MPD Thrust Derivation and Consistency Check

This section derives the electromagnetic thrust produced by the AF-MPD acceleration stage described in Sections A2.4 and A3.1, and checks whether the resulting exhaust velocity and power are internally consistent with the ve = 50 km/s design point used elsewhere in this paper. This check is independent of, and does not modify, the energy-balance (A3.4), Lawson (A3.5), or magnetic-confinement (A3.6) analyses; it reuses their numerical outputs only as inputs, as noted explicitly below.

A3.8.1 Force model

The electromagnetic body force on the current-carrying plasma is, exactly:

$$F = \int (J \times B)\, dV \tag{16}$$

For a first-order engineering estimate, the plasma column is approximated as carrying a uniform discharge current I over an effective current-carrying length L, threading a uniform applied field of magnitude B at angle θ to the current direction. This is NOT an exact description of the plasma (which has nonuniform J and B, finite divergence, and complex current paths); it is a lumped-parameter approximation standard in MPD-thruster engineering analysis [5]:

$$F \approx I\, L\, B \sin(\theta) \tag{17}$$

where I is the discharge/plasma current (A), L is the effective current-carrying length (m), B is the applied magnetic-field magnitude (T), and θ is the angle between current and field (θ = 90° for the perpendicular, force-maximizing geometry assumed here as an engineering default).

A3.8.2 Electromagnetic coupling factor

Real plasma current density and magnetic-field distributions are nonuniform, the plasma diverges rather than forming an ideal column, and some current paths (e.g., near-electrode return currents) do not contribute directly to axial thrust. These effects are captured with an effective electromagnetic thrust-coupling factor ηF, defined as the ratio of the actual axial electromagnetic force to the idealized F = ILB sin(θ) estimate:

$$F_{actual} = \eta_F \cdot I\, L\, B \sin(\theta) \tag{18}$$

ηF is not a generic "loss factor" but a specific measure of how effectively the applied current and field geometry converts into useful axial thrust, encompassing current-density nonuniformity, field nonuniformity, plasma divergence half-angle, and non-axial current paths. Neither this paper's underlying concept notes nor the AF-MPD reference cited elsewhere in this paper [5] provide a validated ηF for this specific geometry, so no definitive value is assigned. Section A3.8.3 instead solves for the ηF required to reproduce the paper's existing performance assumptions, and Section A3.8.4 evaluates that required value for physical plausibility, rather than assuming a preliminary range (e.g., 0.5-1.0) without justification.

A3.8.3 Exhaust velocity and thrust: predicted vs. existing

The electromagnetic force couples to the exhaust stream through momentum flux:

$$F = \dot{m} \cdot v_e \implies v_e = F/\dot{m} \tag{19}$$

A mass flow rate is not independently stated in the underlying concept notes. Consistent with the burn-duration bracket already adopted in Section A3.4 [2], $\dot{m}$ =

mGa/t_burn using the 65 t baseline gives $\dot{m} \approx 0.752$ kg/s (1-day burn) to $\approx 8.36\times10^{-3}$ kg/s (90-day burn).

The only I and B values established anywhere in this paper are the Z-pinch confinement values from Section A3.6 (I ≈ 10.0 MA, B ≈ 401 T), which were derived for the fusion-ignition pinch geometry, not independently for the AF-MPD acceleration stage of Section A2.4. No separate AF-MPD-stage current or field is given. Using these values with the illustrative pinch length L ≈ 0.05 m (Section A3.5/A3.6 geometry assumption) as the only available proxy for an effective conductor length, and θ = 90°, the idealized (ηF = 1) force is:

$$F_{ideal} = I\,L\,B \approx (1.00\times10^{7}\ \text{A})(0.05\ \text{m})(401\ \text{T}) \approx 2.01\times10^{8}\ \text{N} \quad (20)$$

This is an upper bound ($\eta F \le 1$). Dividing by the mass flow rates above gives the AF-MPD-model-predicted exhaust velocity:

$$v_{e,predicted} = F_{ideal} / \dot{m} \approx 2.67\times10^{8}\ \text{m/s (89\% of c), 1-day burn;}\ \approx 2.40\times10^{10}\ \text{m/s (>80× c), 90-day burn} \quad (21)$$

Both results are unphysical (the 90-day case exceeds the speed of light by more than an order of magnitude; the 1-day case is a substantial fraction of c, invalidating the non-relativistic momentum treatment used here). This is reported directly, per the zero-fabrication standard, rather than adjusted. It indicates that the Z-pinch confinement I and B (Section A3.6) cannot be the actual continuous, or pulse-averaged, electromagnetic acceleration parameters acting on the gallium exhaust — those confinement values describe a brief (~1 μs, Section A3.5), extreme-magnitude pulse needed momentarily for fusion ignition, not a sustained or repetitively-averaged accelerating force on a much larger, slower mass flow. Treating them as interchangeable is not physically justified and is the direct cause of the inconsistency found here.

Conversely, solving Eq. 18-19 for the ηF required to reproduce the existing ve = 50 km/s design point, given $F_{existing} = \dot{m}\cdot v_e$:

$$F_{existing} = \dot{m}\cdot v_e \approx 3.76\times10^{4}\ \text{N (1-day)}, \approx 4.18\times10^{2}\ \text{N (90-day)} \quad (22)$$

$$\eta F_{,required} = F_{existing} / F_{ideal} \approx 1.87\times10^{-4}\ \text{(1-day)}, \approx 2.08\times10^{-6}\ \text{(90-day)} \quad (23)$$

These required ηF values (on the order of 10^{-4}-10^{-6}) are far below any physically reasonable range for an electromagnetic coupling factor, which by construction cannot exceed 1 and would not plausibly fall below roughly 0.1-0.5 for a functioning MPD-class thruster [5]. An implausibly small ηF is not a valid resolution; it is a symptom confirming the same conclusion as Eq. 21: the Section A3.6 pinch-confinement I and B are not an appropriate stand-in for the AF-MPD acceleration-stage parameters, and this paper does not currently establish the latter independently.

[Flagged inconsistency: the AF-MPD thrust predicted from the only I, B values available in this paper (Section A3.6, Z-pinch confinement stage) is inconsistent with the existing ve = 50 km/s / mass-flow assumptions (Sections A3.1, A3.4) by 4-6 orders of magnitude, and the exhaust-velocity prediction is non-physical (exceeds c). This does not necessarily invalidate the ve = 50 km/s design point itself (which is independently supported by comparison to published MIF concepts, Section A3.1), but it does mean the AF-MPD electromagnetic force has not been validated against that design point from first principles. Insert Data Here: independently specified AF-MPD acceleration-stage current, field, and effective length, distinct from the Z-pinch confinement parameters, from dedicated circuit/field design or simulation.]

A3.8.4 Energy consistency check

Using the existing ve = 50 km/s and the mass flow $\dot{m}$ derived above (not the non-physical predicted ve from Eq. 21), the kinetic power is:

$$P_{kinetic} = \tfrac{1}{2}\dot{m}v_e^2 = \tfrac{1}{2}F\cdot v_e \approx 9.40\times10^{8}\ \text{W (1-day)}, \approx 1.04\times10^{7}\ \text{W (90-day)} \quad (24)$$

This matches $P_{useful,avg}$ already computed in Section A3.4 exactly, because both use the same $\dot{m}$, ve, and burn-duration bracket; this is a consistency self-check, not a second independent energy model, and confirms that the existing ve/$\dot{m}$/P_{useful} triple (Sections A3.1-A3.4) is internally self-consistent on its own terms. It does not resolve the Eq. 20-23 discrepancy, which concerns whether the Section A3.6 electromagnetic parameters can actually produce that ve/$\dot{m}$ pair from first principles.

A3.8.5 Thrust efficiency

The propulsive/thrust efficiency of the AF-MPD stage specifically is:

$$\eta_{thrust} = P_{kinetic} / P_{input} \quad (25)$$

where P_{input} here must mean only the electrical power delivered to the AF-MPD current-drive and field circuit — the portion of the Section A3.4 loss chain labeled "electromagnetic acceleration losses" and "nozzle/exhaust losses" (Section A3.4, items v-vi) — not the total fusion power P_{fusion}, and not the overall η_{total} defined in Section A3.4. η_{total} already spans the entire chain from fusion yield to useful exhaust kinetic energy (heating/vaporization/ionization, burn fraction, neutron loss, radiation loss, EM coupling, and nozzle losses combined); η_{thrust} as defined here would be only the EM-coupling/nozzle sub-portion of that chain, isolated from the upstream fusion-related losses. Computing η_{thrust} against η_{total} would double-count losses already inside η_{total}. This paper does not separately quantify a stand-alone AF-MPD electrical input power distinct from the aggregate η_{total} treatment in Section A3.4:

[PARAMETER NOT ESTABLISHED — REQUIRES SOURCE OR SIMULATION: AF-MPD-stage electrical input power P_{input}, and therefore η_{thrust}, cannot be numerically evaluated with the information available in this paper.]

A3.8.6 Parameter sensitivity

For the simplified model at fixed θ = 90°, thrust scales linearly with each of current, field, and effective length, and with the coupling factor:

$$F \propto \eta F \cdot I \cdot B \cdot L \quad (26)$$

i.e., $\partial F/\partial I = \eta F\ L\ B$, $\partial F/\partial B = \eta F\ I\ L$, $\partial F/\partial L = \eta F\ I\ B$, and $\partial F/\partial \eta F = I\ L\ B$: doubling any single factor doubles F, holding the others fixed, and F is independent of $\dot{m}$ (which instead sets ve = F/$\dot{m}$ for a given F). This is a scaling relationship only, not evidence that thrust can be increased arbitrarily by increasing I or B. Practical ceilings identified elsewhere in this paper constrain all four factors simultaneously: electrode current loading and ablation-driven erosion limits on I and current density (Section A2.3); magnetic-pressure structural stress, found in Section A3.6 to already exceed candidate electrode material strength by 60-200× at the Section A3.6 field level ($\sigma B \approx 64$ GPa); thermal loading from neutron and (partially deposited) alpha flux (Section A3.5); and the pulsed-power driver energy and repetition-rate constraints identified in Section A3.4. Any parameter combination proposed to close the Eq. 20-23 discrepancy must be checked against these limits, not chosen in isolation.

A3.8.7 Consistency table

Table A2. AF-MPD parameter consistency check.

Parameter	Existing value	Required/derived	Status
Exhaust velocity ve	50 km/s (A3.1)	Pred.(ηF=1): 2.67×10^{8}-2.40×10^{10} m/s	INCONSISTENT (>c)
Mass flow $\dot{m}$	[NOT STATED]	0.752-8.36×10^{-3} kg/s (derived, A3.4 bracket)	DERIVED, not sourced
Current I	10.0 MA (A3.6, pinch)	Not est. for AF-MPD stage	AMBIGUOUS
Field B	401 T (A3.6, pinch)	Not est. for AF-MPD stage	AMBIGUOUS
Cond. length L	[NOT STATED]	0.05 m (borrowed, A3.5/A3.6)	ASSUMPTION
Thrust F	[NOT STATED]	37.6 kN-418 N (ve,$\dot{m}$); 2.01×10^{8} N (Lorentz)	INCONSISTENT (4-6 OoM)
Kinetic power Pk	= Sec.A3.4 Puseful	9.40×10^{8}-1.04×10^{7} W	CONSISTENT
Input power P_in	[NOT STATED]	cannot be derived	NEEDS SOURCE/SIM
Thrust eff. ηthrust	[NOT STATED]	cannot compute w/o P_in	NEEDS SOURCE/SIM
Req. ηF (match ve)	-	1.87×10^{-4} (1d); 2.08×10^{-6} (90d)	IMPLAUSIBLY LOW

A3.8.8 Scientific status of this section

(A) Derived directly from the proposed equations in this section: Fideal (Eq. 20), ve,predicted (Eq. 21), Fexisting and ηF,required (Eq. 22-23), Pkinetic (Eq. 24), and the $F \propto \eta F\ I\ B\ L$ scaling (Eq. 26).

(B) Adopted from the existing paper: ve = 50 km/s (Sec. A3.1); mGa = 65,000 kg and the 1-90 day burn-duration bracket (Sec. A3.4); I ≈ 10.0 MA and B ≈ 401 T (Sec. A3.6, pinch-confinement stage, reused here only for lack of a dedicated AF-MPD-stage value); L ≈ 0.05 m (Sec. A3.5/A3.6 geometry assumption).

(C) Literature-derived: the lumped $F \approx ILB\sin(\theta)$ approximation as a standard first-order MPD-thruster engineering model [5].

(D) Engineering assumptions made in this section: $\theta = 90°$ (perpendicular, force-maximizing geometry); reuse of the Section A3.6 pinch I, B and Section A3.5/A3.6 geometry L as proxies for AF-MPD-stage values, explicitly flagged as unvalidated for this purpose.

(E) Requires future simulation or experiment: a dedicated AF-MPD-stage current, field, and conductor-length specification distinct from the Z-pinch confinement parameters; the AF-MPD electrical input power Pinput and resulting ηthrust; and the electromagnetic coupling factor ηF itself. A first-order Lorentz-force calculation of the kind performed here does not validate the complete AF-MPD plasma dynamics; a full validation would require, at minimum, an MHD/plasma simulation resolving current-density distribution, magnetic-field distribution, plasma flow, and electromagnetic coupling in the actual thruster geometry.

Net finding: the AF-MPD thrust model, evaluated with the only current and field values available in this paper (the Section A3.6 Z-pinch confinement parameters), is not consistent with the existing ve = 50 km/s / mass-flow assumptions used elsewhere in this paper; the discrepancy is large (orders of magnitude in force, and a non-physical exhaust-velocity prediction) and is reported here rather than concealed. This does not overturn the paper's existing conclusions, which already treat ve = 50 km/s as a literature-comparison-supported design point rather than a first-principles result (Section A3.1), but it does add a specific, previously unidentified gap — a dedicated AF-MPD-stage electromagnetic specification, independent of the fusion-confinement parameters — to the follow-on work already recommended in Section A9.

A4. Engineering Constraints and Design Limits

A4.1 Capillary Pore Sizing (Young–Laplace Analysis)

The porous electrode structure must retain liquid gallium against surface tension under passive (unpumped) conditions while still permitting capillary-driven flow to the ablating surface during operation. The Young–Laplace relation gives the pressure differential ΔP sustainable across a curved liquid interface of radius r:

$$\Delta P = 2\gamma / r \quad (4)$$

Using the experimentally measured surface tension of liquid gallium at its melting point, $\gamma = 0.708$ N/m (verified against sessile-drop measurements reported in the materials-science literature [7]), and a target hold pressure of $\Delta P \approx 1$ atm (1×10^{5} Pa) as a conservative design margin against zero-gravity leakage, the maximum permissible pore radius is:

$$r = 2\gamma / \Delta P = 2(0.708) / (1\times10^{5}) \approx 1.42\times10^{-5}\ \text{m} \approx 14\ \mu\text{m} \quad (5)$$

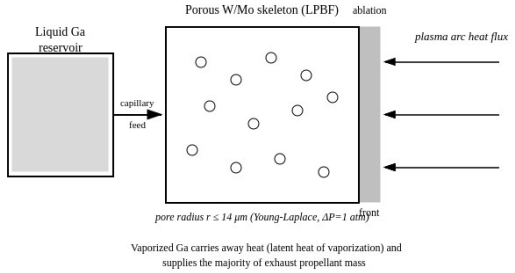


Fig. A5. CPS electrode cross-section: capillary gallium feed through the porous refractory skeleton to the plasma-facing ablation front.

Standard Laser Powder Bed Fusion processes typically achieve minimum feature and pore resolutions on the order of 20–30 μm [8]. The required pore radius therefore sits at or below the practical resolution limit of current LPBF manufacturing — a substantive, currently unresolved constraint on the concept's manufacturability. Closing this gap would likely require either (i) a dedicated post-processing step (e.g., partial sintering, infiltration, or chemical pore refinement) to reduce as-printed pore size below the as-built LPBF resolution, or (ii) relaxation of the passive-retention design requirement in favor of active propellant management (e.g., pressurized or pump-fed gallium delivery), trading the design's passive-retention simplicity for additional moving parts and failure modes.

A4.2 Neutron Transport and Electrode Degradation

D-T fusion neutrons are emitted at 14.1 MeV. Because gallium has a comparatively low macroscopic neutron cross-section at this energy, only a small fraction of the 14.1 MeV neutron flux is expected to be absorbed within a sweating layer on the order of 100–500 μm thick; the layer is expected to meaningfully attenuate the alpha-particle and soft X-ray heat load associated with the pinch, but not the fast-neutron flux.

An order-of-magnitude estimate of the neutron mean free path in liquid gallium can be obtained from $\lambda = 1/(n\sigma)$, where n is the atomic number density of liquid gallium ($n \approx \rho N_A/M \approx 5.3\times10^{28}$ m^{-3}, using the measured liquid density $\rho \approx 6.10$ g/cm³ near the melting point [9]) and σ is the total microscopic cross-section at 14 MeV.

[Flagged placeholder: a total neutron cross-section of σ ≈ 2 barns was used for an illustrative mean-free-path estimate (λ ≈ 9–10 cm); partial (n,2n), (n,p), and (n,n'α) cross-section data for gallium near 14 MeV are reported in the nuclear-data literature [10], but a verified total cross-section could not be confirmed against a nuclear data library (e.g., ENDF/B-VIII.0 or JENDL-5) within the scope of this paper — Insert Data Here: verified total neutron cross-section for natural gallium at 14.1 MeV. The resulting mean free path should be treated as an order-of-magnitude estimate only.]

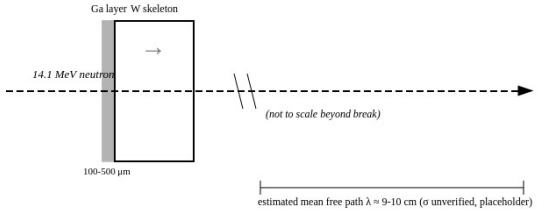


Fig. A6. Neutron mean-free-path schematic (not to scale): the thin Ga layer attenuates thermal/charged-particle flux but not the fast-neutron flux reaching the W skeleton.

Even at this order-of-magnitude level, the estimated mean free path (several centimeters) is one to two orders of magnitude larger than the sweating-layer thickness, indicating qualitatively that the gallium layer's protective role is limited to thermal and charged-particle shielding rather than neutron shielding. The practical consequence is that the underlying tungsten (or molybdenum) skeleton is expected to remain exposed to the full 14.1 MeV neutron flux over the engine's operating life. Fast-neutron irradiation of tungsten is documented in the fusion first-wall materials literature to drive transmutation, principally to rhenium and osmium via neutron capture and subsequent beta decay, along with associated radiation-induced embrittlement and microstructural degradation [11]. Over a multi-burn mission profile, this transmutation and embrittlement process represents a credible electrode lifetime constraint that is not addressed by the gallium ablation-cooling mechanism and should be assessed independently, ideally through a dedicated neutron-transport and materials-damage simulation using verified cross-section libraries.

A4.3 Gallium Oxide Passivation

Liquid gallium forms a rigid, sub-nanometer-to-few-nanometer Ga_2O_3 oxide skin within milliseconds of exposure to trace oxygen or moisture, a well-documented property exploited elsewhere in gallium-based liquid-metal engineering (e.g., shape-retaining soft-robotic and reconfigurable-electronics applications) [12]. In the context of a capillary-fed electrode, this same passivation mechanism poses a risk of pore choking, since residual outgassing of spacecraft materials (H_2O, CO_2) and any atomic oxygen entrained from prior operation in low Earth orbit could progressively seal the capillary network.

The concept notes propose two mitigations: alloying the gallium with active wetting agents (e.g., titanium or magnesium) to disrupt oxide-layer formation, or periodically flushing the capillary network with hydrogen plasma to reduce the oxide back to metallic gallium.

[Flagged placeholder: neither anti-passivation mitigation has been demonstrated for this specific capillary-electrode application in the sources reviewed for this paper — Insert Data Here: experimental validation of anti-passivation strategy.]

A5. Operational Risk Factors

A5.1 Plume Backflow and Spacecraft Contamination

All plasma thrusters exhibit some degree of plume backflow, in which a small fraction of exhaust particles

diverge from the primary thrust axis and travel back toward the spacecraft body due to magnetic field divergence and charge-exchange collisions. For gaseous propellants, backflow particles are typically volatile and do not accumulate. Gallium, by contrast, condenses readily on contact with any surface below its 2,204°C boiling point — essentially any spacecraft surface. Backflow-driven metallic film accumulation on solar arrays, optical sensors, thermal radiators, and exposed electronics is therefore a credible contamination risk specific to a condensable liquid-metal propellant, and would need to be bounded through plume-divergence modeling and, likely, physical baffling or standoff geometry between the thruster and sensitive spacecraft surfaces before flight qualification.

A5.2 Tritium Fuel-Cycle Constraint

Tritium is not naturally abundant and decays with a 12.3-year half-life, precluding long-term stockpiling. Civilian tritium supply is dominated by Canada's fleet of CANDU heavy-water reactors, which collectively produce on the order of 2 kg of tritium per year as a byproduct of reactor operation, against a global civilian stockpile estimated at approximately 25–30 kg; these figures are broadly consistent across multiple recent policy and technical sources reviewed for this paper [13].

Under the (unverified, lower-bound) 0.24 kg D-T fuel-mass estimate developed in Section A3.3, and assuming a D-T mixture split evenly by mole between deuterium and tritium, a single engine burn could require on the order of a tenth of a kilogram of tritium — a non-trivial fraction of annual global civilian tritium production. This demand would compete directly with the tritium requirements of terrestrial fusion research programs, including ITER, and would require a dedicated, likely government-controlled, fuel-procurement and handling pathway rather than commercial off-the-shelf sourcing.

A5.3 Liquid-Metal Slosh and Attitude Control Interaction

A propellant load on the order of tens of tonnes of liquid gallium (density ≈ 6.1 g/cm³, roughly six times that of water) represents a substantial, mobile mass fraction of the vehicle. In microgravity, an unbaffled liquid mass of this scale is susceptible to slosh dynamics that can couple destructively with the spacecraft's attitude control system during maneuvers, particularly given gallium's high density relative to conventional cryogenic propellants. This is a well-characterized class of problem in spacecraft propellant-tank design, typically addressed through slosh baffles, compartmentalized tankage, or slosh-mass dynamic models integrated into the attitude control system design. The concept notes do not yet specify a tank or baffle architecture; a coupled slosh–attitude-control analysis is recommended as a design task rather than treated as resolved.

A6. Regulatory and Planetary Protection Considerations

A6.1 Treaty Framework for Pulsed Nuclear Propulsion

The 1963 Partial (Limited) Test Ban Treaty prohibits nuclear weapon test explosions in the atmosphere, in outer space, and underwater, while the 1967 Outer Space Treaty's Article IV prohibits placing nuclear weapons or other weapons of mass destruction in orbit or on celestial bodies. Neither treaty was drafted with pulsed fusion propulsion specifically in mind, and whether a device that produces thrust through repeated small-scale fusion pulses would be classified as a prohibited nuclear test or weapon, as opposed to a propulsion system, is not settled by the treaty text itself [14,15].

This ambiguity is not new: it was a significant factor, alongside the 1963 treaty's atmospheric and space testing provisions, in the termination of Project Orion, the 1950s–1960s nuclear-pulse propulsion program [14], and it remains an open legal question for any pulsed nuclear propulsion concept, including the one considered here. Resolving it would require a dedicated legal and policy analysis beyond the scope of this engineering paper; the concept should not be assumed to be legally unproblematic on the basis of the physics analysis alone.

A6.2 Planetary Protection

Neutron activation of the gallium propellant during the fusion pulse could render the exhaust plume mildly radioactive, depending on the specific activation products and half-lives involved, which are not quantified in this paper. COSPAR planetary protection policy restricts the introduction of radioactive or biological contamination to solar-system bodies of interest for prebiotic chemistry or the search for life, including Mars [16]. A mission profile that involves directing an activated exhaust plume toward, or into the upper atmosphere of, such a target body would need to demonstrate compliance with these restrictions, likely through activation-product inventory modeling and plume-dispersion analysis, before mission approval could be expected.

A6.3 Launch Authorization

In the United States, the launch of any spacecraft containing a space nuclear system is governed by National Security Presidential Memorandum 20 (NSPM-20, issued August 20, 2019), which established a risk-informed safety analysis and launch-authorization process administered through the Interagency Nuclear Safety Review Board [17]. Comparable systems (radioisotope power sources, in past NASA missions) have required safety-review processes spanning several years and substantial dedicated budget, independent of the underlying spacecraft's own development schedule. A pulsed-fusion propulsion system carrying tritium and generating a neutron flux during operation should be expected to fall under an equivalent, and likely more extensive, review process; this represents a major programmatic and schedule driver that should be planned for from an early design phase rather than treated as a late-stage formality.

A7. Candidate Mission Profiles

The milli-g-class thrust levels reported for comparable published magneto-inertial fusion propulsion concepts (Section A3.1) imply that this class of engine is best suited to missions that can tolerate a sustained, low-thrust acceleration profile rather than a short impulsive burn. Combined with the unresolved contamination, neutron-shielding, and regulatory questions identified in Sections A4–A6, the most plausible near-to-mid-term applications are uncrewed rather than crewed:

- High-Δv outer-planet orbiter delivery (e.g., a Uranus or Neptune orbiter). Sustained high-Isp thrust could reduce trip time and propellant mass relative to the chemical-propulsion-plus-gravity-assist trajectories used by historical outer-planet missions, at the cost of requiring low-thrust trajectory optimization rather than impulsive maneuver planning.
- Uncrewed bulk cargo pre-positioning in support of a crewed Mars campaign architecture. Routing high-Δv, radiation- and contamination-sensitive propulsion technology through cargo-only vehicles decouples it from the human-safety-critical path while still reducing the overall chemical-propellant mass required for a crewed campaign.
- A fast outer-solar-system or heliopause precursor probe, benefiting from sustained low-thrust acceleration over a multi-year cruise phase, in the spirit of prior electric-propulsion precursor mission concepts but with a potentially higher achievable cumulative Δv.

Mission Profile	Regime	Key Driver	Crew Status
Outer-planet orbiter delivery (e.g. Uranus/Neptune)	Sustained low-thrust, multi-year cruise	High Isp reduces trip time / propellant vs. chem+gravity-assist	Uncrewed
Bulk cargo pre-positioning	Low-thrust, non-time-critical	Decouples novel propulsion from crew safety path	Uncrewed
Heliopause / outer-system precursor	Continuous thrust, multi-year cruise	High cumulative Δv over long cruise	Uncrewed

Fig. A7. Candidate mission profiles compared by thrust regime, key performance driver, and crew status.

Crewed applications are deferred pending resolution of the contamination, neutron-shielding-mass, tritium-supply, and regulatory questions identified in Sections A4–A6; none of these are dismissed as unsolvable, but none are yet resolved to a level that would support a credible crewed mission design.

A8. Supporting Infrastructure Requirements

Maturing this concept toward flight readiness would require ground-based infrastructure substantially beyond what is needed for a conventional electric or chemical propulsion system:

- A secure, likely government-coordinated tritium supply chain and handling facility, given the scarcity of civilian tritium and its competing use in terrestrial fusion research programs (Section A5.2).
- Specialized Laser Powder Bed Fusion manufacturing and non-destructive evaluation capability qualified for sub-20-μm porous refractory-metal structures, or an equivalent post-processing capability to reach the pore sizes required in Section A4.1.
- Ground-based pulsed-power and plasma test facilities capable of simulating Z-pinch-relevant current and neutron environments, comparable in class to existing national-laboratory pulsed-power facilities used for Z-pinch and inertial confinement fusion research.
- Radiation-hardened power management and avionics for the pulsed, high-current drive electronics, and for any onboard systems located near the neutron-emitting engine.
- On-orbit assembly and fueling infrastructure, given the large absolute propellant tank volumes implied by a multi-tonne gallium load, even accounting for gallium's favorable density relative to cryogenic propellants.
- A nuclear launch safety review and authorization pathway (e.g., the NSPM-20 / Interagency Nuclear Safety Review Board process in the U.S. context, or an equivalent framework in other launching states), understood as a multi-year, resource-intensive programmatic requirement rather than a technical footnote (Section A6.3).
- A planetary protection compliance review, particularly for any mission profile that would direct an exhaust plume toward a body of astrobiological interest (Section A6.2).

A9. Conclusion

This paper has formalized the operating principles of a gallium-fueled fusion-choke AF-MPD thruster concept and evaluated its performance envelope against standard propulsion and plasma-physics relations. The concept's assumed exhaust velocity range is broadly consistent with values reported for comparable published magneto-inertial fusion propulsion studies, and the associated propellant mass-fraction reduction relative to chemical propulsion is substantial in an idealized, impulsive-Δv comparison.

At the same time, this analysis has identified several specific points in the underlying concept notes that require further work before the concept's feasibility can be credibly assessed: an unstated fusion-to-propellant energy coupling efficiency that renders the fuel-mass estimate a lower bound rather than a design figure; an unreconciled discrepancy between two independently assumed propellant-mass scenarios; an unverified neutron cross-section value underlying the electrode neutron-shielding assessment; a capillary pore-size requirement at or below current additive-manufacturing resolution; and open questions regarding spacecraft contamination, tritium supply, liquid-metal slosh dynamics, and treaty-level regulatory status.

Recommended follow-on work includes: (i) energy-balance modeling using a realistically bounded fusion-to-propellant coupling efficiency rather than an idealized 100% conversion assumption; (ii) verification of the gallium neutron cross-section against a formal nuclear data library; (iii) reconciliation of the vehicle propellant-mass budget across a single, internally consistent set of assumptions; and (iv) replacement of the impulsive rocket-equation estimate with a low-thrust trajectory optimization appropriate to the

concept's milli-g-class acceleration regime. None of the constraints identified here appear to be fundamental physical impossibilities; each instead defines a specific, tractable area of further engineering and materials-science study.

References

[1] J. Miernik et al., “Z-Pinch fusion-based nuclear propulsion,” Acta Astronautica, vol. 82, p. 173, 2013.

[2] J.T. Slough, “The Fusion Driven Rocket,” in Proc. 2024 IEEE Aerospace Conference, 2024.

[3] T. Auger, V. Michel, L. Cassayre, et al., “Liquid metal embrittlement and deformation induced martensite: the case of 316L austenitic steel LME by liquid eutectic gallium-indium,” Corrosion Science, 2021.

[4] M. Tajmar et al., “High current liquid metal ion source using porous tungsten multiemitters,” Ultramicroscopy, 2010.

[5] B. Wang et al., “A 100 kW Class Applied-Field Magnetoplasmadynamic Thruster,” Journal of Visualized Experiments, 2018.

[6] J. Cassibry et al., “Case and Development Path for Fusion Propulsion,” Journal of Spacecraft and Rockets, 2015.

[7] S.C. Hardy, “The surface tension of liquid gallium,” Journal of Crystal Growth, vol. 71, no. 3, pp. 602–606, 1985.

[8] A. Talignani et al., “A review on additive manufacturing of refractory tungsten and tungsten alloys,” Additive Manufacturing, 2022.

[9] V.P. Chentsov, V.G. Shevchenko, A.G. Mozgovoi, M.A. Pokrasin, “Density and surface tension of heavy liquid-metal coolants: Gallium and indium,” Inorganic Materials: Applied Research, 2011.

[10] J. Luo, R. Liu, L. Jiang, “Cross section measurements for gallium in the neutron energy range of 13.5 to 14.8 MeV,” Radiochimica Acta, 2012.

[11] A. Hasegawa et al., “Neutron Irradiation Behavior of Tungsten,” Materials Transactions, 2013.

[12] J.-H. Kim, S. Kim, M.D. Dickey, J.-H. So, H.-J. Koo, “Interface of gallium-based liquid metals: oxide skin, wetting, and applications,” Nanoscale Horizons, vol. 9, no. 7, pp. 1099–1119, 2024.

[13] R.J. Pearson, A.B. Antoniazzi, W.J. Nuttall, “Tritium supply and use: a key issue for the development of nuclear fusion energy,” Fusion Engineering and Design, vol. 136, pp. 1140–1148, 2018.

[14] G.R. Schmidt, J.A. Bonometti, C.A. Irvine, “Project Orion and Future Prospects for Nuclear Pulse Propulsion,” Journal of Propulsion and Power, vol. 18, no. 3, pp. 497–504, 2002.

[15] G.H. Reynolds, J.L. Outten, “Pulsed Nuclear Space Propulsion and International Law: Some Preliminary Observations,” Journal of Air Law and Commerce, vol. 87, no. 3, p. 445, 2022.

[16] K. Olsson-Francis et al., “The COSPAR Planetary Protection Policy for robotic missions to Mars: a review of current scientific knowledge and future perspectives,” Life Sciences in Space Research, vol. 36, pp. 27–35, 2023.

[17] National Security Presidential Memorandum 20 (NSPM-20), “Launch of Spacecraft Containing Space Nuclear Systems,” The White House, Washington, DC, 20 August 2019.

Bibliographic fields (volume, issue, page) are included only where independently confirmed; otherwise omitted rather than estimated, per this manuscript's zero-fabrication standard.